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Review

Mitochondria at the crossroads of health and disease

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SUMMARY

Mitochondria reside at the crossroads of catabolic and anabolic metabolism—the essence of life. How their structure and function are dynamically tuned in response to tissue-specific needs for energy, growth repair, and renewal is being increasingly understood. Mitochondria respond to intrinsic and extrinsic stresses and can alter cell and organismal function by inducing metabolic signaling within cells and to distal cells and tissues. Here, we review how the centrality of mitochondrial functions manifests in health and a broad spectrum of diseases and aging.

INTRODUCTION

Mitochondria are efficient, adaptable bioenergetic machines that power eukaryotic cells biosynthetically. Mitochondria display incredible functional and compositional diversity even within an organism, despite being derived from a common bacterial ancestor, predicted to be of alpha-proteobacterial origin. The endosymbiotic event that gave rise to mitochondria was pivotal in the evolution of eukaryotic cells. Endosymbiosis fundamentally altered cellular organization—mitochondria became one of only two endomembrane systems in addition to the endoplasmic reticulum (ER) in non-photosynthetic eukaryotes that contain a genome. Direct and intimate interactions between these two major endomembrane systems evolved into critical organizational and regulatory hubs in cells.

Mitochondria lost most of the bacterial ancestral genome, and many of these genes were transferred to the nucleus: current eukaryotic genomes carry over 600 genes of alpha-proteobacterial origin.[1] The vast majority (99%) of the mitochondrial proteome in humans—over 1,100 proteins—is encoded by the nuclear genome. The small 16.6-kb plasmid-like mitochondrial genome (mtDNA) encodes only 13 proteins devoted to producing the chemical energy molecule ATP via oxidative phosphorylation. This reduced size of mtDNA enables its dramatic amplification in cells. The mitochondrial-nuclear partnership and the dispersed and efficient mitochondrial compartment configuration enabled cells to boost their energy production and dramatically increase nuclear gene expression capacity by powering transcription and protein translation—the highest consumers of cellular ATP. This allowed eukaryotic cells to evolve into relatively large and functionally diverse entities.[2] Loss of the ancestral genome also drove the evolution of new pathways for mitochondrial protein and lipid biogenesis and mitochondrial behavior to integrate their function with the cell state.

The centrality of mitochondrial function for eukaryotic cells is reflected in how their dysfunctions manifest in a spectrum of diseases that span almost all medical fields. Remarkably, over 40% of mitochondrial proteins are linked to human diseases, underscoring the centrality of mitochondria to human health.[3] Mitochondrial diseases present at any age, in any organ system, in a highly tissue-specific or multisystemic manner and display any inheritance pattern.[4] How this essential organelle causes such diverse diseases is still largely a mystery. For example, mutations of complex I of the respiratory chain, the first enzyme of the ATP-producing oxidative phosphorylation system, can cause isolated optic atrophy in young adults[5] or subacute necrotic brain disease in children (Leigh disease). A defect in mitochondrial translation can cause strokes early in life, hearing loss, and diabetes or hypertrophic cardiac disease.[4] mtDNA depletion syndrome can manifest either in the skeletal muscle or as a brain-liver disease but not elsewhere.[6,7] These examples indicate that the underlying mechanisms of mitochondrial diseases extend far beyond deficits in ATP production. The physiological complexity of mitochondrial diseases also highlights the importance of understanding the multifaceted nature of mitochondria in a cellular context. True leaps in knowledge have been made possible by transformative technologies, such as metabolomics and proteomics, illuminating the significant effects of mitochondria and their diseases on cellular and systemic metabolic homeostasis.

Structurally, mitochondria harbor two distinct membranes: the outer and inner membranes (OM and IM, respectively) and two functionally diverse compartments: the intermembrane space (IMS) and the matrix (Figure 1). The OM is a smooth lipid-enriched surface containing ancestral beta-barrel proteins,

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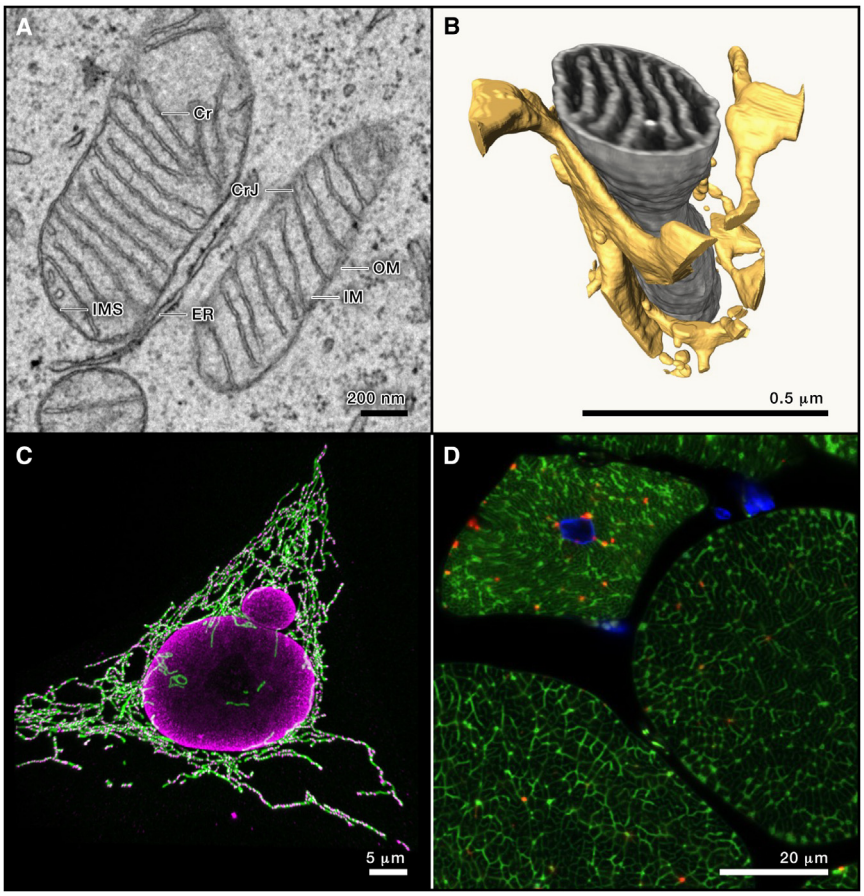


Figure 1. Mitochondrial morphology: Compartmentalized, interactive, dynamic, and mobile

(A) Transmission electron micrograph of human embryonic kidney cell line. IM: inner membrane; OM, outer membrane; IMS, intermembrane space; Cr, cristae; CrJ, Crista junction; ER, endoplasmic reticulum. Scale bar, 200 nm.

(B) Focused ion beam scanning electron micrograph of a mitochondrion and endoplasmic reticulum, depicting their close interactions. Scale bar, 0.5 μm.

(C) Immunofluorescence microscopy of U2OS cell. Mitochondrial network (green) and DNA (magenta) depicting mtDNA nucleoids within a fused mitochondria network. Scale bar, 5 μm.

(D) Mitochondrial networks *in vivo*, in the skeletal muscle of a mouse expressing *mito*-QC reporter; mitochondria (green), nuclei (blue), mitolysosomes (red). Scale bar, 20 μm. Image credit: Eija Jokitalo and Ilya Belevich (A and B), Juan Cruz Landoni (C), and Takayuki Mito (D).

including voltage-dependent anion-selective channels (VDACs)/porins, which permeabilize the membrane to molecules up to 10 kDa in size. The OM also contains alpha-helical membrane proteins, primarily with only one transmembrane domain, such as MFF and MIRO,[8] which are receptors for the mitochondrial division dynamin and motility machinery, respectively, underscoring the role of the OM in the control of mitochondrial behavior and cellular communication.[9] The IMS is the smallest compartment, housing only 5% of the mitochondrial proteome. However, the IMS is critical as it facilitates diverse pathways for protein and lipid biogenesis, metal homeostasis, redox regulation and oxidative folding, apoptosis, and mitochondrial dynamics and structure.[10] The IM is the most protein-dense membrane in eukaryotic cells, highly impermeant and shaped into structures called cristae. The IM is recognized as the functional core of the organelle—it houses the electron transport chain, also called the respiratory chain, with respiratory enzyme complexes (RCs; in humans, complex I–IV) and a turbine-like ATP synthase machine (RC V) that dimerizes to shape and drive the formation of cristae.[11,12] The mitochondrial matrix houses a vast number of biosynthetic reactions as well as the mitochondrial genetic system. It is the home of the tricarboxylic acid cycle (TCA) that oxidizes the carbon backbones of nutrients to NADH- and FADH2-reducing equivalents, which are used by the RCs to generate chemiosmotic energy for ATP synthesis, as well as by many other biosynthetic pathways. This compartmentalization is essential for overall organelle and cellular function.

The discovery of molecular players and mechanisms underlying primary mitochondrial diseases is essential for understanding common diseases linked to mitochondrial alterations, such as Parkinson's disease, retinopathy, and diabetes. The progression of different types of cancers is also affected by mitochondrial metabolism. Many of these diseases are age dependent, consistent with the observations that mitochondrial functions are altered with aging in a tissue-specific manner. Such alterations could result from a chronic shift toward mitochondrial biosynthetic repair functions at the cost of ATP production. This impacts the cell state through changes in transcription and translation and cell-extrinsic factors, such as the extracellular matrix. As such, mitochondria have been recognized as contributors to most age-related degenerative diseases. Furthermore, altered metabolism can also cause adverse drug effects. Here, we provide an update on advances in understanding the mechanistic basis of mitochondrial functions and structure. We highlight their crucial roles in regulating the balance of anabolism and catabolism in cells and organisms and how this is relevant for pathophysiology.

MITOCHONDRIA AS METABOLIC HUBS

Metabolic programs are critical in determining cell state by providing the necessary energy and molecular precursors for various cellular functions and influencing gene expression and signaling pathways. Mitochondria are at the heart of setting such metabolic programs as they either house or are connected to all intermediary metabolic pathways, dynamically changing their metabolic mode based on nutrient, stress, and differentiation status of cells.

Multifunctional metabolic organelles

The best-known cellular role of mitochondria is the production of ATP via oxidative phosphorylation. RCs I, III, and IV form the electron transport chain, transferring reducing equivalents from glycolysis and the TCA cycle to the final electron acceptor, inhaled oxygen, which produces CO_2 and H_2O.[12] The RCs can function as solitary units, but they also assemble into defined supercomplexes, thought to increase the efficiency of respiration, although these are not essential for life.[13,14] Supercomplexes also confer stability to respiratory complexes, which may be important during stress. Electron transfer by RCs is coupled to proton transport, which generates a tremendous electrochemical potential ($\Delta\Psi$mv) of 150–200 mV across the IM. From the viewpoint of a molecule, this $\Delta\Psi$mv is equivalent to the field strength of a lightning bolt and is used to power the rotary turbine-shaped engine, complex V, ATP synthase, to catalyze the formation of ATP from ADP and Pi.

Mitochondrial $\Delta\Psi$mv functions in many non-ATP-related aspects of mitochondrial biology. For example, $\Delta\Psi$mv powers and regulates mitochondrial ion homeostasis and the production of reactive oxygen species (ROS), which act as physiological and pathological signaling molecules.[15] $\Delta\Psi$mv is also essential for mitochondrial fusion and protein import. In this context, $\Delta\Psi$mv has been proposed to function as a sentinel for mitochondrial health. Mitochondrial dysfunction within the network is detected by a decrease in $\Delta\Psi$mv. Decreases in $\Delta\Psi$mv impede the import of nuclear-encoded mitochondrial proteins that, in turn, initiate cellular stress responses. Examples of such pathways are the mitochondrial integrated stress response (ISRmt), which metabolically buffers the cell to promote recovery, and PINK1-mediated mitophagy, which initiates the degradation of mitochondria that are beyond repair. Loss of $\Delta\Psi$mv also occurs in cell death via the opening of the mitochondrial permeability transition pore (mtPTP) and BCL associated X (BAX)-mediated cytochrome *c* release during apoptosis. Under conditions where oxidative phosphorylation is disrupted, such as during hypoxia, ATP synthase can run in the reverse direction to maintain $\Delta\Psi$m at the expense of the hydrolysis of ATP supplied by glycolysis, underscoring its regulation and importance.

Mitochondrial metabolic plasticity

Although mitochondria retain some ancestral functions and responses, such as the ability to respond to viral infections and temperature,[16] there are significant changes in their metabolic roles and plasticity. They synthesize iron-sulfur (Fe/S) clusters, an essential cofactor for RC enzymes, DNA repair, and redox-sensitive enzymes.[17] Mitochondria take part in synthesizing the indispensable heme-cofactor, utilized in oxygen binding, redox reactions, drug detoxification, and management of ROS, as well as the regulation of transcription and translation. Regulatory steps of cholesterol biosynthesis occur in mitochondria. Mitochondria also produce central regulatory metabolites such as acetyl-CoA, hydroxy-ketoglutarate, 2-oxoglutarate, succinate, and fumarate that modify the epigenome to tune nuclear gene expression and function as signaling organelles, mediating the metabolic state to the whole organism (Figure 2A).

On the cellular level, mitochondria respond directly and via signaling pathways to nutrient state—a critical determinant of cell state. During fasting, mitochondria transition to catabolic oxidative burners of all available fuels: glucose, lipids, amino acids, and ketone bodies. When nutrients are plentiful, ATP is produced via non-oxygen-dependent glycolysis, which facilitates the transformation of mitochondria to an anabolic mode. In the anabolic mode, mitochondria utilize nutrient carbons to drive the production of amino acids, nucleotides, and other metabolites via the TCA and folate cycles to promote growth and repair (Figure 2B).[18] This mode prevails in the proliferating state of many cell types to support building-block synthesis, including cancer cells (known as the Warburg effect), different tissue progenitor cells, and primed pluripotent stem cells. Switching metabolic programs to upregulate mitochondrial oxidative phosphorylation promotes differentiation and functional specification of pluripotent cells. The control of cellular fate by mitochondrial metabolism is a burgeoning area in mitochondrial biology.

Pyruvate is a critical regulatory metabolite at the crossroads of cytoplasmic glycolysis and mitochondrial oxidative phosphorylation, which is important, for example, for stem cell quiescence.[19] Its utilization by mitochondria is tightly controlled by the activities of the mitochondrial pyruvate carrier, as well as by kinases and a phosphatase, which respectively inactivate and activate pyruvate dehydrogenase, which converts pyruvate to the hub metabolite acetyl-CoA to fuel the TCA cycle.[20] In some somatic stem cells, however, the primary mitochondrial mode is fatty acid oxidation (FAO), which maintains quiescence or is required for activation to proliferate.[21] Why quiescent stem cells are strongly dependent on FAO is currently unclear. FAO could provide cells with a redox state required for maintaining quiescence or protect them against the build-up of toxic metabolic intermediates. An increased reliance on glycolysis reduces the self-renewal capacity of adult stem cells,[22] relevant for an aging-related altered metabolic program. Understanding how various regulatory enzymes direct carbon fluxes in different cell types under physiological and pathophysiological states is essential for new intervention targets to buffer disease states and age-related changes.

Mechanistically, mitochondrial metabolic plasticity is linked to structural plasticity.[23] Mitochondrial shape and distribution result from the concerted actions of the mitochondrial division, fusion, and motility machines. In glycolytic pluripotent embryonic stem cells, mitochondria are fragmented and underdeveloped in their internal cristae structure. As stem cells differentiate, mitochondria undergo concerted changes: they switch to oxidative phosphorylation for energy production and smaller elongated mitochondria and form networks with extensive cristae. These changes in mitochondrial structure are partly determined by the relative rates of mitochondrial division and fusion, catalyzed by dynamin-related GTPases DRP1 in the cytosol and MFN1/2 and OPA1, in the mitochondrial OMs and IMs, respectively.[23] The activities of these membrane-shaping proteins are required in cell fate transitions and integrated with metabolism.[24] The organization and localization of biosynthetic enzymes within and near mitochondria are also regulated by the cellular metabolic state. For example, inside the organelle, enzymes in the coenzyme Q (CoQ) biosynthetic pathway organize into precisely positioned supramolecular metabolon.[25] Similarly, *de novo* purine synthesis pathway

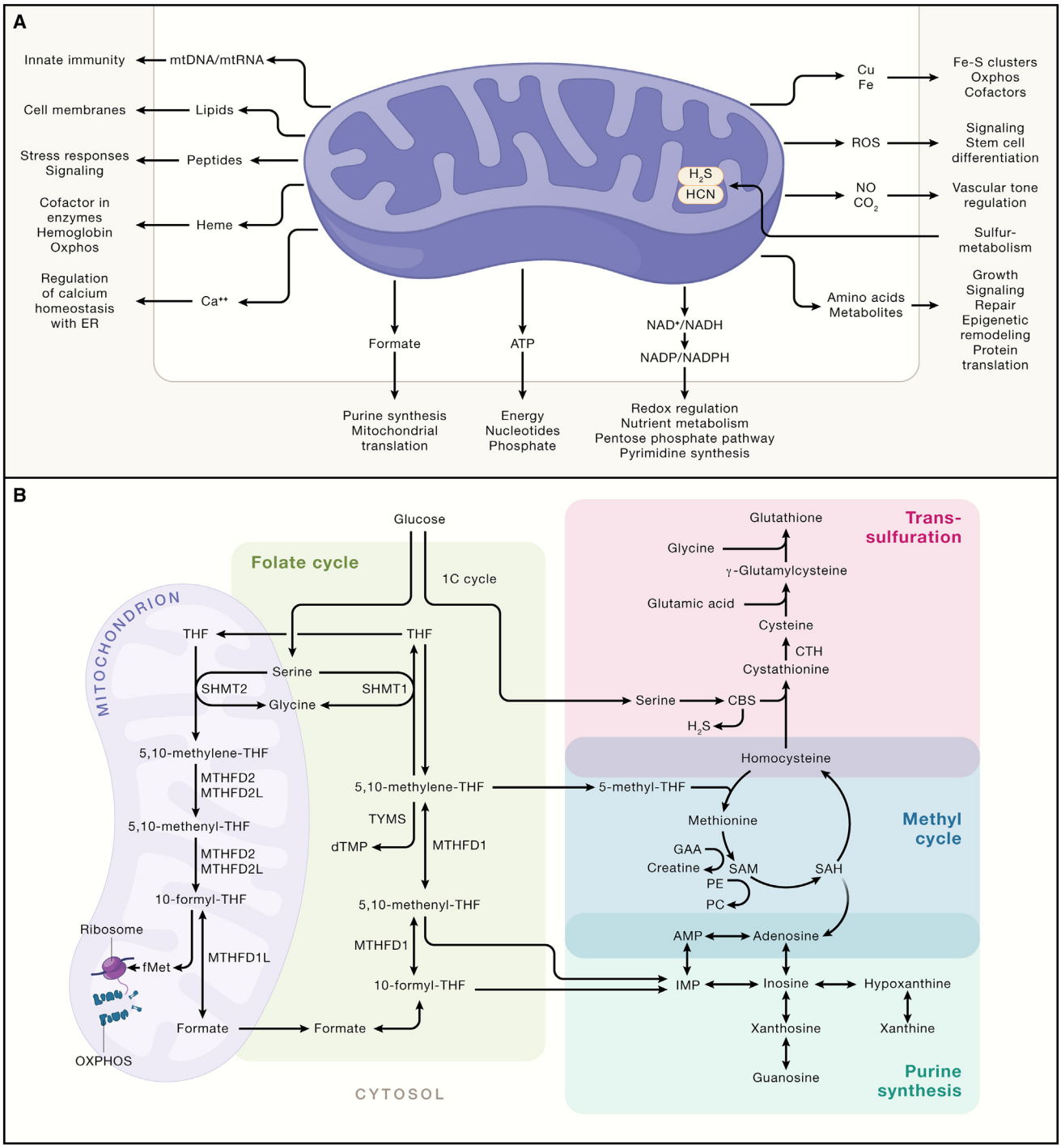


Figure 2. Multifaceted mitochondrial contributions to cellular homeostasis

(A) Mitochondria produce cofactors and metabolites to cells and tissues, as well as releasing them as systemic signals to tune metabolism. Mitochondria are also a storage site for metals and contain high amounts of B-vitamin derivatives (folates, NAD-forms) and the redox-regulator glutathione.

(B) Mitochondrial-cytoplasmic folate- and one-carbon cycle. Mitochondria have an essential role in providing forms of folate for the major biosynthetic, anabolic pathways that provide growth components for the whole cell.

enzymes cluster on the OM to form a purinosome metabolon that channels mitochondrial-derived formate into purine synthesis.[25,26] Thus, dynamic changes in mitochondrial structure are not simply products of metabolic changes. Understanding the mechanistic links between metabolism and mitochondrial behavior is an important active research area.

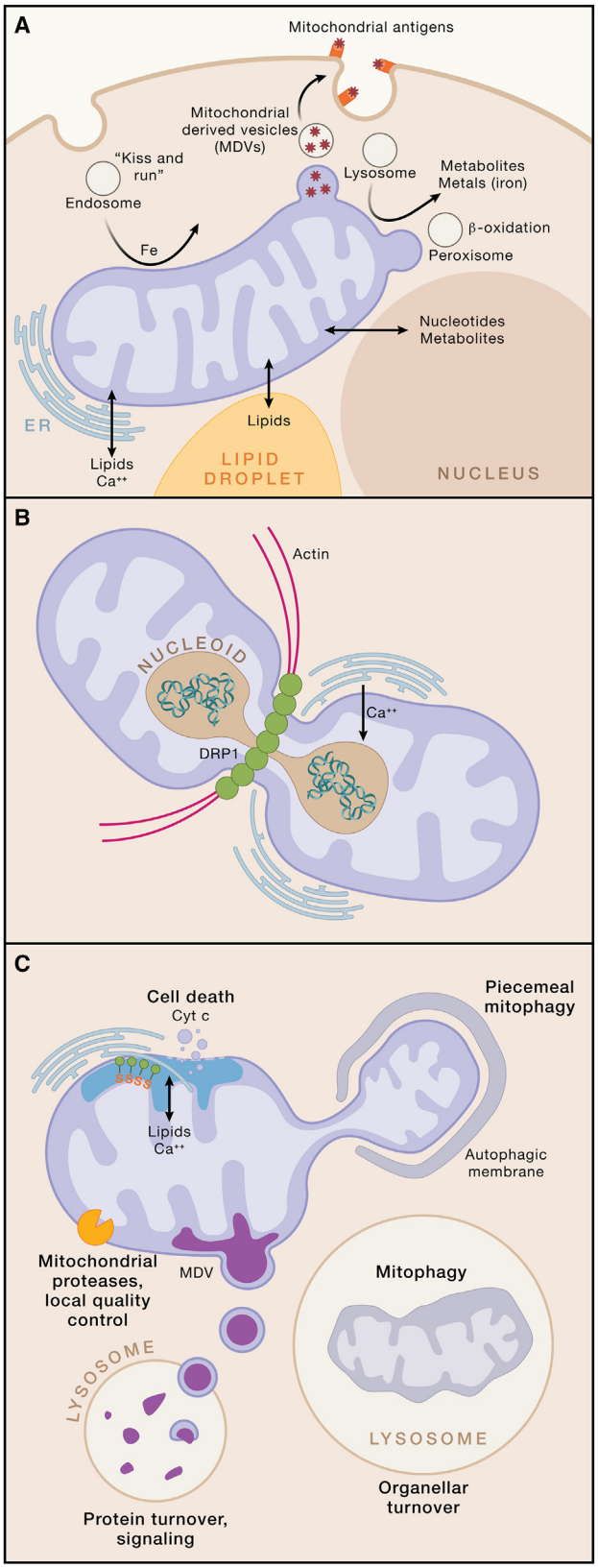


Figure 3. Mitochondrial interactions with organelles

(A) Mitochondrial contacts with cellular organelles. ER, endoplasmic reticulum; E, endosome; L, lysosome; P, peroxisome; LD, lipid droplet; MDVs, mitochondrial-derived vesicles.

(B) Mitochondrial division site is marked by a replicating mtDNA nucleoid at an ER-contact site and likely regulated by ER-derived calcium signaling. DRP1 (dynamin-related protein 1, green circles) is recruited to the contact site and, together with cytoskeletal actin (red filaments), constricts mitochondria to divide.

(C) Examples of different events involved in mitochondrial quality control, turnover of mitochondrial contents, or leading to death. Sumoylated (S) DRP1 (green circles) constricts mitochondrial membrane releasing cytochrome *c* (cyt *c*) from intermembrane space and inducing cell death. The event involves ER-derived calcium and lipid signaling. MDVs contain damaged proteins delivered to lysosomes. Mitophagy: selective autophagy turning over mitochondria. Blue membrane: loss of membrane potential. Lilac membrane: low membrane potential and budding MDVs.

Beyond identifying dynamic changes in mitochondria, the field has started to describe distinct "mitotypes" within a cell. Mitochondrial sub-populations exist, for example, in brown fat cells, where mitochondria-contacting lipid droplets are specialized for lipid storage, whereas unassociated mitochondria are active in oxidative phosphorylation.[27] Indeed, mitochondrial specialization may more broadly be mediated by contacts with other cellular organelles, such as ER. Mitochondrial contact sites mediate the exchange of lipids, ions, and metabolites and regulate mitochondrial dynamics, mtDNA replication, and quality control (Figure 3). They are formed in a manner responsive to cell and metabolic state and likely fine-tune the output of mitochondria in ways we do not yet understand. The diversity of mitochondrial composition, organization, and physiology at the cellular level highlights their essential roles in the evolution of eukaryotic organisms.

In multicellular organisms, mitochondria have evolved to adapt their functions to the specific needs of different cell types. However, how mitochondria match their biosynthetic functions with the metabolic requirements of the tissue has yet to be fully understood, and how these decisions contribute to specific organismal vulnerabilities is another open question. For example, it is a mystery how the heart, which is highly dependent on ATP, maintains metabolic homeostasis in most mitochondrial diseases and how highly anabolic cells, such as oligodendrocytes, prioritize glucose carbon flux to promote their specific biosynthetic phospholipid pathways.

Interestingly, adjacent cells with distinct characteristic steady-state metabolic profiles can share metabolites. For example, the stem cell niche has anabolic "caretaker" cells that provide metabolites for stem cells. The caretakers are biosynthetic and use glucose as their main fuel, while the stem cells are considered oxidative (Figure 4).[28] Mitochondrial functions also depend on and dictate cellular differentiation, organ maturation state, and organismal sex, age, and environment.

The context is also important: a genetic defect in a mitochondrial ribosome (mitoribosome) subunit often causes a severe cardiac disease that is typically lethal in early life. However, if the child survives the first few years, the heart condition can stabilize, negating a mortality risk.

These examples highlight the gaps in knowledge of mitochondrial tissue-, age-, and context-specific roles that are

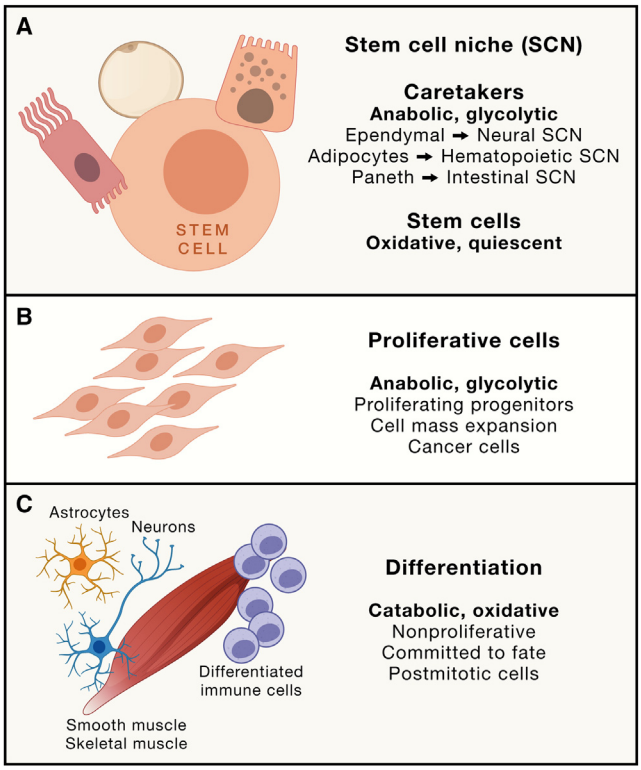


Figure 4. Metabolic modes of cells in their proliferative and non-dividing states

(A) The stem cell niche has anabolic "caretaker" cells that provide metabolites for stem cells. Ependymal cells support neural stem cells, adipocytes support hematopoietic stem cells, and Paneth cells are caretakers of the intestinal stem cells. The caretakers are anabolic and use glucose as their main fuel, while the stem cells are considered oxidative.

(B) Proliferating cells mainly use glucose as their fuel and also shuttle glucose carbons to anabolic biosynthesis for synthesis of cell membrane, amino acid, and nucleotide components.

(C) Shift to oxygen using catabolic metabolism is an early event in differentiation processes, associated with mitochondrial mass increase in all cell types, including astrocytes, neurons, non-proliferating epithelia, and white blood cells.

highly relevant for understanding and mitigating disease. Metabolic signaling at the level of the organism and its intimate link to dietary metabolism are also critical. For example, depletion of a mitochondrial protease in both muscle and heart or high-fat feeding can suppress the severe cardiac phenotypes resulting from the loss of the same protease in the heart.[29]

Emerging knowledge indicates that even when mitochondrial disease phenotypes are apparent in a single tissue, they broadly affect the organism through systemic metabolism and hormone-like signaling.[30–32] For example, in muscle mitochondrial disease, muscle-derived "metabokines" travel systemically to modify glucose uptake and mitochondrial biogenesis in a specific region of the brain.[33] There is a need to determine how mitochondrial-controlled metabolism changes with different stages of human diseases and/or the maturation of postnatal tissue and aging. In addition, a more detailed understanding of the organism-wide manifestation of mitochondrial diseases and their underlying molecular and cellular mechanisms is critical for a better understanding of basic biology and physiology. High-resolution precision tools for spatial, cell-specific analyses in intact tissues currently being developed provide increasing insights into these questions.

Mitochondrial roles in folate cycle and anabolic responses

Mitochondria have essential functions in the main anabolic biosynthesis pathway—the folate-driven one-carbon (1C) cycle[34] (Figure 2B). 1C-metabolic pathway enzymes are distributed between mitochondria and the cytoplasm, with some folate-modifying enzymes residing in the nucleus. Mammals derive folate, vitamin B9, from their diet with a minor component provided by the gut microbiota. Folate is the main carrier of 1C-units to growth-related biosynthesis pathways and is a perfect candidate to signal food availability at the cellular level. Folic acid is converted into active tetrahydrofolate forms (THFs) in the liver and released into circulation. THFs enter cells via specific folate receptors, where they are polyglutaminylated via mitochondrial or cytoplasmic folylpoly-gamma-glutamine-synthetase enzymes, resulting in a pool of stable compartment-specific THFs in postmitotic cells. Different folate forms are highly specific and rate-limiting for the enzymes and pathways they serve. For example, methylene-THF drives the methyl cycle, and mitochondrial-derived formate feeds the cellular purine synthesis via 10-formyl-THF and formyl-methionine generation, required for mitochondrial translation initiation, to facilitate organellar protein synthesis.[35] The mitochondrial folate cycle produces NADPH and NADH that, together with mitochondrial aspartate, are required for pyrimidine synthesis.[34]

NAD+ and glutathione: Redox-regulators of mitochondrial and cellular metabolism

Mitochondria also produce reduction-oxidation (redox) regulators—metabolites that direct metabolic flux and biosynthesis pathways, serve as antioxidants, modify proteins, and maintain redox homeostasis. The main regulators are NAD forms and glutathiones: NAD+, NADH, NADP, NADPH, GSH, and GSSG. The redox-active pathways that produce these regulators are linked, as GSSG is recycled back to GSH by NADPH-dependent enzyme glutathione reductase.[36] NAD^+ is a critical nucleotide cofactor for the TCA cycle, which produces NADH as the main carrier of nutrient-derived electrons from the cycle to complex I of the respiratory chain, which regenerates NAD+. Recently, the mitochondrial NAD+ transporter SLC25A51 was described,[37,38] emphasizing the compartment-specific functions of cytoplasmic and mitochondrial NAD+ pools. The different forms of vitamin B3 (nicotinic acid, nicotinamide, nicotinamide mononucleotide, and nicotinamide riboside) are NAD+ precursors driving synthesis through tissue-specific enzymatic pathways. NAD metabolites broadly regulate over 200 cellular biosynthetic pathways and the direction of their flux. They utilize NAD+ as an electron acceptor, and NAD+ and its forms also function as cofactors. For example, NAD+ is a cofactor for the NAD+-dependent protein lysine deacetylases that regulate various cellular functions, including nutrient sensing, energy metabolism, and stress responses.[39] NAD+ pools have wide-ranging effects on cellular functions and homeostasis.

Mitochondria are the main regulators of cellular and systemic NAD+/NADH ratio, and through this function, mitochondrial status broadly influences cell and organismal health.

Our understanding of the relevance of NAD+ metabolism for human health and diseases is increasing. In mice and humans with mitochondrial muscle disease, NAD+ pools and 1C metabolism are altered, and NAD+ deficiency is observed in the muscle and blood as a secondary consequence of disease.[40] Importantly, restoration of NAD+ pools with NAD+ precursors in these patients corrected NAD+ levels and defects in mitochondrial structure and function and slowed disease progression.[40] The patients showed remarkable whole-body benefits including reduced liver fat and increased muscle mass, strength, and oxidative phosphorylation activity. This restoration in function shifted metabolism toward healthy controls but did not correct the causative genetic defect or the associated transcriptional stress responses. In addition to mitochondrial muscle disease, NAD+ deficiency and the consequential coupled metabolic alterations are increasingly being linked to various degenerative, aging-related diseases of humans, including neurodegeneration, cancer cachexia, and cardiomyopathy.[41–43] These observations highlight the need to evaluate whether restored NAD+ levels may act as a buffer to treat a broad range of human diseases and/or increase human health span.

Glutathione and its oxidized and reduced forms, GSSG and GSH, represent the second major redox modifier and antioxidant.[44] While glutathione is not synthesized in mitochondria, the mitochondrial folate cycle is closely linked to the transsulfuration pathway, responsible for producing cysteine, hydrogen sulfide (H_2S), and glutathione. Like NAD+:NADH, the GSH:GSSG ratio is a critical measure of the cellular redox state, which controls key cellular processes, including cell proliferation, differentiation, and cell death. The half-life of glutathione is short—only hours—but its cellular pools reach millimolar concentrations, and up to 15% of glutathione exists in mitochondria, sequestered by SLC25A39- and SLC25A40-dependent transport.[45–47] GSSG is recycled back to its reduced form through the NADPH-dependent enzyme glutathione reductase. Glutaredoxin enzymes harness the reducing power of GSH to regulate redox homeostasis and act as a defense system against oxidative stress. The GSH/glutaredoxin system is also required for *de novo* dNTP synthesis by ribonucleotide reductase to maintain nucleotide pools and mtDNA replication in postmitotic cells. Ferroptosis is a form of iron-dependent, mitochondrial-linked cell death associated with decreased GSH biosynthesis and the consequent inactivation of the glutathione peroxidase, GPX4, resulting in lipid peroxidation and cell death.[48] In this manner, dNTP production competes with GPX4-dependent lipid detoxification to use GSH, linking GSH metabolism to ferroptosis sensitivity. Increased flux toward GSH synthesis also produces H_2S, a potent complex IV inhibitor that is degraded by mitochondria, providing a mitochondrial feedback loop.[49]

Glutathione usage is high in the nervous system and decreased levels of GSH are associated with the progression of neurodegenerative disorders. In neurons, the production capacity of GSH is limited and is significantly supported by astrocytes, which fuel neuronal GSH synthesis by providing cysteine and glutamate. As such, GSH metabolism is considered a therapeutic target of neurodegenerative disorders: the glutathione precursor N-acetyl-L-cysteine is clinically used to rescue liver function in response to acetaminophen intoxication, but its delivery to the brain is controversial.[47]

Reactive oxygen species

ROS are common side products of oxidative phosphorylation. Their regulation is important because of their potential to cause oxidative damage to DNA, proteins, and lipids. They also act as signaling molecules to regulate cell state such as, for example, muscle growth, stem cell differentiation, and erythrocyte maturation.[50,51] Mitochondria possess multiple mechanisms for generating ROS. Mitochondrial complexes I and III are predominant ROS producers, generating O_{2-}, which reacts with H_2O to produce H_2O_2.[52] Complex I can also generate O_{2-} by reverse electron transfer under conditions where the pool of reduced coenzyme-Q-pool and proton motive force are relatively high. Under these conditions, at complex I, electrons from $CoQH_2$ are transferred back to NAD+, with an associated increase in succinate. Conditions that favor reverse electron transfer include ischemia-reperfusion. Recent data suggest that succinate acts as a signal for the reduction state of the Q-pool and controls superoxide production from complex I, performing as a redox-active metabolite. Succinate can also be secreted and regulate ATP production, oxygen sensing, and hypoxia signaling.[53] These functions highlight the roles of systemic metabolite signaling initiated by mitochondrial metabolism. ROS signaling and ROS sensitivity are highly tissue-specific, as exemplified by the actions of the antioxidant MitoQ, a modified form of CoQ10, which accumulates in mitochondria in a manner dependent on membrane potential. MitoQ doses that mitigate defects in erythrocyte precursors are toxic for neural stem cells in mice with increased mtDNA mutagenesis.[51] Furthermore, antioxidants inhibit physical training-related muscle growth. Thus, like many other signals, there is a defined range of efficacy beyond which lies toxicity that can manifest in a tissue-specific manner.

BUILDING AND SHAPING MITOCHONDRIA

The composition and regulation of the mitochondrial proteome and lipidome are highly tissue- and context-specific to tailor function to need.[54] For example, electron transport chain proteins are more abundant in the heart and skeletal muscle to meet the high energetic needs for muscle contraction. Cutting-edge mass spectrometry approaches have been used to determine mitochondrial protein and lipid compositions, representing an incredible advancement for the field. These mitochondrial atlases have been extensively mined using genetics, multi-omics, and bioinformatics to give deep insights into mitochondrial biology.[3,55–57]

Diverse and dynamic mitochondrial proteomes and lipidomes

The human mitochondrial proteome amounts to about 7% of the total cellular proteome, containing proteins with an incredible range of abundances (over six orders of magnitude) and half-lives (from minutes to months), providing important clues for their function and regulation. Mass-spectrometry-based approaches

have resolved the composition of mitochondrial sub-compartments and how proteins interact within them.[3,54] Most mitochondrial proteins are localized to the IM and matrix and dedicated to metabolism and functioning. For example, they are directly involved in oxidative phosphorylation or indirectly by mediating the expression of the handful of mitochondrial genes essential for respiration. A full annotation of how mitochondrial proteomes differ between proliferative and postmitotic cells, between different tissues and cell types and even between mitochondria within a cell, is still needed. Such atlases could help identify additional mitochondrial protein functions that are linked to disease.

Like the proteome, the lipidome is complex, membrane-specific, and dynamic, with over 200 species, including the predominant mitochondrial glycerolipids, phosphatidylcholine (PC) and phosphatidylethanolamine (PE), and the unique signature lipid, cardiolipin (CL), inherited from its bacterial origins.[58] Mitochondria also contain small amounts of sphingolipids and sterols, except mitochondria in steroid-hormone-synthesizing cells. In the liver, the mitochondrial lipidome oscillates in a diurnal, likely circadian-directed manner in an opposite phase to the nuclear lipidome, highlighting that cellular lipid fluxes are critical regulators of cellular and organismal homeostasis.[57] Understanding mitochondrial lipid regulation in different cell types and how they more broadly impact cellular and organismal lipid homeostasis will reveal new biology and therapeutics. Indeed, peptides that stabilize mitochondrial CLs are already in clinical trials for mitochondrial diseases.

Nuclear-encoded mitochondrial protein biogenesis

Nuclear-encoded mitochondrial protein biogenesis is initiated close to the mitochondrial OM on cytosolic ribosomes, which can also be membrane-tethered to enable efficient protein import, sorting, and assembly.[59] Molecular chaperones play a critical role in the post-translational import of mitochondrial preproteins by maintaining an unfolded state and guiding them to the mitochondrial OM import machinery. The cytosolic ubiquitin-proteosome system (UPS) prevents the accumulation of mitochondrial precursor proteins in the cytosol and acts as an important homeostatic regulator of mitochondrial biogenesis. In the OM, the translocase of the outer membrane (TOM) complex is the gateway for protein import and contains the pore-forming beta-barrel protein Tom40, which does the work of protein import. In humans, MTCH1/2 also function in the OM as direct protein insertases for single and rare multi-pass alpha-helical membrane proteins synthesized in the cytosol.[60] Beyond the TOM complex, the routes and translocation mechanisms for proteins vary based on their sub-mitochondrial destinations and their diverse structural, chemical, and folding features.[61,62] For example, β-barrel proteins are inserted from the IMS by the concerted actions of the small translocase of the inner membrane (TIM) proteins and the sorting and assembly machinery (SAM) complex in the OM. Mitochondrial protein import across the IM depends on the TIM23 translocase that forms a channel that allows proteins to be pulled into the matrix by the mitochondrial Hsp70 motor complex. The IM $\Delta\Psi$mv is also required for precursor import and acts to aid import in part via electrophoretic effects on the relatively basic mitochondrial import signals. In this context, import signals vary in terms of their charge properties. Thus, conditions that change membrane potential can affect protein import efficiency in a substrate-selective manner over a large dynamic range. How membrane potential fine-tunes the mitochondrial proteome in different contexts is not fully understood and is important for fully elucidating mitochondrial plasticity and disease mechanisms.

Under conditions in which the capacity of import and UPS turnover mechanisms is exceeded, cytosolic precursor proteins accumulate and import can arrest—a condition termed mitochondrial precursor overaccumulation stress (mPOS).[63] Multiple cellular stress pathways are activated by mPOS, including pathways that upregulate proteostasis and the mitochondrial ISRmt and mitoCPR (mitochondrial compromised protein import response), mitophagy, and cell death.[63] Recently, a large E3 ligase complex was identified, termed silencing factor of the integrated stress response (SIFI), that resolves import stress by degrading both cytosolic precursors and the key transducers of the ISRmt, DELE1, and the eIF2-α kinase heme-regulated inhibitor (HRI).[64] SIFI analysis indicates its function in concluding stress response signaling is necessary for cellular survival. In this context, SIFI is mutated in ataxia and early-onset dementia, suggesting that chronic stress signaling is a causative factor of these diseases and potentially any disease associated with defective import. Consistently, genetic defects in mitochondrial protein import result in neuropathies, myopathies, neurodegenerative diseases, cancer, and cardiovascular disorders.[4]

Mitochondrial-encoded protein biogenesis

In humans, the 16.6-kb circular mtDNA contains 37 genes: 22 tRNAs and two ribosomal RNAs for mitochondrial translation of the 13 structural proteins encoded by mtDNA, which are core components of the respiratory chain protein complexes I, III, IV, and V.[65] mtDNA is transcribed as a polycistronic transcript by mitochondrial RNA polymerase, and mRNAs are typically flanked by tRNA genes recognized by specific processing enzymes, which release and process the ends of the mRNAs and tRNAs.[66] The transcription machinery proceeds through the first transcripts-tRNA-Phe, 12S rRNA, tRNA-Val, and 16S rRNA—and after completion of this operon, transcription frequently terminates instead of proceeding through the entire polycistronic transcript. This ensures the appropriate stoichiometry and abundance of mitoribosome structural RNA components to enable translation of the 13 mitochondrially encoded proteins.[66] The entire process of mitochondrial gene expression is interconnected, giving rise to RNA granule condensates that can be visualized using super-resolution microscopy.[67,68] Most of the mitochondrially encoded proteins are highly hydrophobic IM proteins. Consequently, they are translated and inserted into the IM by mitoribosomes associated with the ancient Oxa1 insertase complex, which promotes the biogenesis of some nuclear-encoded IM proteins.[69]

Significant recent advances in cryo-electron microscopy have captured mitoribosome structures, including from humans.[70,71] Human structures reveal that in addition to mitochondrial-encoded 16S and 12S rRNAs, the ribosome possesses a third integral RNA component: tRNA-Val, encoded in mtDNA and present in equal stoichiometry. Within the mitoribosome, tRNA-Val forms extensive interactions with mitoribosomal proteins as a structural replacement for the bacterial 5S rRNA. Mitoribosome

structures have also provided many mechanistic insights into tRNA recognition, polypeptide synthesis, the roles of ribosome modifications, and the co-translational mode of insertion of mitochondrial-encoded proteins in the IM.[72] The mitoribosome is modified by metabolites that are in other contexts known to represent different metabolic states. The role of such modifications to mitoribosomal functions remain to be studied. Understanding ribosome structures also allows explanations for clinical manifestations of mitochondrial diseases caused by mitoribosomal defects.

The nuclear-encoded and mitochondrial-encoded translation products converge on the biogenesis of the I, III, IV, and V respiratory complexes. How this is coordinated is an area of active research. In yeast, cytosolic translation controls mitochondrial translation to facilitate balanced mitochondrial and cytosolic protein synthesis.[73] In contrast, in human cells, imbalance caused by inhibiting one system does not affect the other in the short term.[74] In humans, the nuclear-encoded respiratory complex assembly component MITRAC15 also functions in the translational regulation of mitochondrial-encoded subunits of complexes I and IV. Miscommunication between the mitochondrial and nuclear genomes can cause activation of stress pathways such as the ISRmt. Thus, more work is needed to identify additional coordination mechanisms and how they contribute to disease and aging.[75]

Mitochondrial translation defects

Since 2010, next-generation exome sequencing (NGS) approaches have revealed pathological mutations in nuclear genes that cause a spectrum of different mitochondrial translation and respiratory complex assembly defects on a cellular level. Mitoribosome subunit protein defects are a common cause of infantile hypertrophic cardiomyopathies, highlighting the importance of mitochondrial biogenesis in the heart after birth. In contrast, phenotypes associated with mitochondrial aminoacyl-tRNA synthetases (ARS2s), also essential for mitochondrial translation, manifest in a highly variable manner in different organ systems at different ages. ARS2s can cause neonatal cardiomyopathy (AARS2), cerebral white matter disease (DARS2, MARS2), encephalopathy and developmental delay (e.g., EARS2, AARS2, FARS2), hearing loss, and ovarian dysfunction (HARS2, LARS2).[76] Based on amino acid abundance, mitochondrial tRNA synthetases can redundantly conjugate multiple similar amino acids. Such events are known to occur, for example, for mammalian AARS2, which can replace serine with alanine tRNA-Ala, which has deleterious consequences for the protein function.[77] In contrast to cytosolic ARS2s, only some mitochondrial ARS2s possess an editing domain to release mischarged amino acids, shifting quality control of mitochondrial protein synthesis to post-translational protease-dependent mechanisms. Examples of variants in ARS2s that cause cellular malfunction only in combination with specific mtDNA variants exist, demonstrating the complexities arising from the dual genetic mitochondrial biogenesis system.[78] These interactions are challenging to identify because of the inherent variability of mtDNA and the common occurrence of nuclear gene variants of unknown significance. Surprisingly, defects affecting mitochondrial translation rarely manifest a respiratory chain deficiency in cultured cells from patients, while defects in the translation products—respiratory chain enzyme subunits—often show a cellular phenotype.

Complex I: A megacomplex challenging to assemble

Complex I of the respiratory chain is a beautiful example of how nuclear and mitochondrially encoded proteins coordinate for assembly. Seven of the 45 subunits of complex I are encoded by mtDNA with the rest of the structural components and at least 14 assembly factors encoded by nuclear genes.[79–81] These are separately translated on mitochondrial and cytoplasmic ribosomes and assembled as modules. The membrane arm of complex I contains all the mtDNA-encoded subunits. A peripheral matrix-localized arm, composed of 17 nuclear-encoded subunits, has two functional parts: a matrix proximal NADH-oxidizing N-module and a membrane adjacent ubiquinone-reducing-Q module. The two arms associate together to form an L-shaped complex with seven Fe/S clusters that form an "iron wire" through the peripheral arm, which accepts electrons from NADH and transfers them to complex III via the electron carrier CoQ. Complex I coordinately pumps four hydrogen atoms via its membrane arm to the IMS. This modular assembly of complex I allows separate control of module turnover and assembly. The N-module, the primary site of ROS production, is degraded at a significantly higher rate than the rest of the enzyme by the matrix-localized ClpXP protease, which is required for optimal complex I function in cells.[82] A similar modular strategy could exist for complex V as well, where decapping the ATP synthase reveals the membrane-embedded c-ring, forming a leak channel and loss of $\Delta\Psi mv$.[83] For complex I, such a modular mechanism might reduce ROS in transient metabolic stress, and for complex V, it might resolve protein import stress. However, the mechanisms require further research to reveal their potential physiological roles.

Complex I deficiencies are a common cause of Leigh syndrome, a fatal infantile-onset disease affecting the brain stem and basal ganglia. Indeed, almost any gene mutation affecting a nuclear-encoded respiratory chain subunit or assembly factor can cause Leigh syndrome.[5] One reported pathological mechanism in mice with complex I accessory subunit (NDUFS4) inactivation involves aberrant hypoxia signaling. This promotes hypervascularization in the affected brain region, flooding the region with too much oxygen and resulting in hyperoxic damage.[84] In the NDUFS4-knockout mice, microglial activation and peripheral lymphocyte invasion also contribute to the local damage.[85]

All complexes have elaborate, stepwise assembly mechanisms, coordinated by specific complex-specific assembly factors, many of which were initially identified from the genetic characterization of mitochondrial diseases[86,87] Their composition may change based on metabolic state (complex II)[88] or tissue-specific functions (complex IV).[89] Their replacement in the IM requires *de novo* synthesis and assembly. However, such turnover occurs slowly in non-dividing cells: the respiratory chain proteins show exceptionally long half-lives to several weeks.[90]

Mitochondrial lipid biogenesis

Like the mitochondrial proteome, the organellar lipidome is made from a combination of extra-mitochondrial and mitochondrial-derived components (Figure 5).[58] The mitochondrial IM

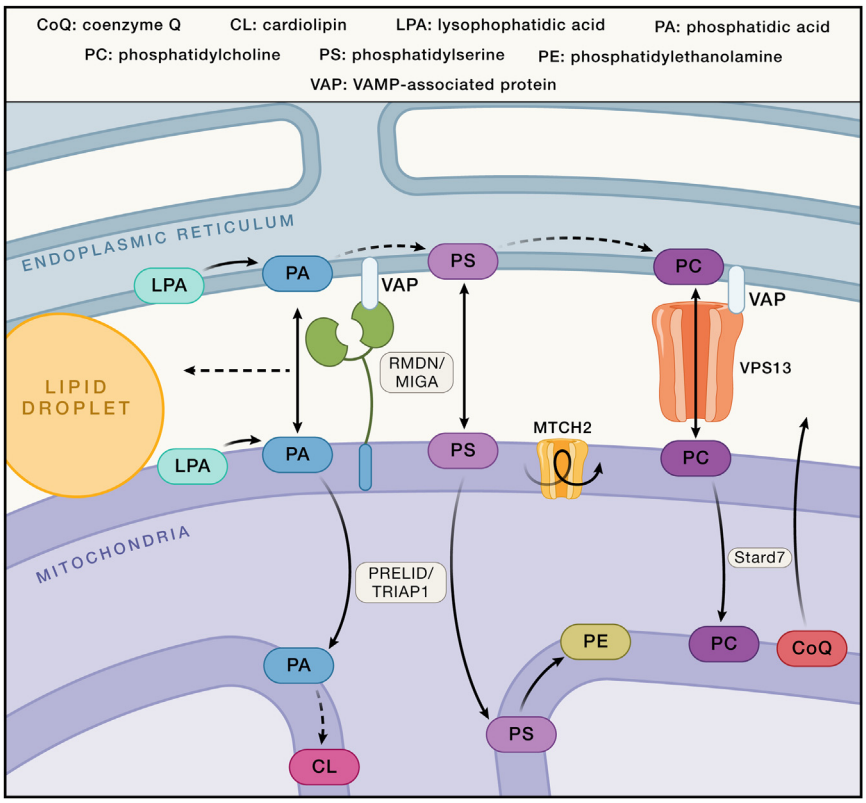


Figure 5. Mitochondrial lipid metabolism
Mitochondria and ER coordinately regulate lipid homeostasis. Precursors of phospholipids enter mitochondria from the ER using specific transport proteins (e.g., RMDN3, MIGA1/2, VPS13A/D, MTCH21/2, PREDLID3b/1, TRIAP1, Stard7). Lipid droplet formation is fueled by the synthesis of di-acyl- and triacyl-glycerides from PA.

possesses essential *de novo* biosynthetic pathways for the glycerolipids PE and CL, as well as for the ancient electron carrier lipid CoQ. The PE and CL pathways are fueled primarily by transporting ER-synthesized precursors, phosphatidic acid (PA) and phosphatidylserine (PS)[91] to the OM. From there, they are shuttled to the IM by the highly conserved human PRELID3b-TRIAP1 and PRELID1-TRIAP1 complexes localized in the IMS, as initially described in yeast.[92] PE is an abundant lipid in OM and IM, whereas CL is highly IM enriched. Together, they play crucial roles in IM structure and organization, being essential regulators of mitochondrial fusion, respiratory complexes integrity, and crista morphogenesis.[91] CL is also in the OM in relatively small amounts, where it recruits the division machinery and BCL proteins to regulate mitochondrial dynamics and apoptosis, respectively.[93] PC is the major phospholipid in mitochondria required for organelle integrity and must be acquired from the ER or other organellar membranes. PC is incorporated into the OM by multiple lipid transport proteins and transferred to IM by the intramitochondrial lipid transfer protein STARD7. In cultured cells, STARD7 is cleaved by the IM peptidase presenilin associated rhomboid-like (PARL), enabling it to operate as an export shuttle for CoQ to the cytosol, where CoQ protects against ferroptosis.[94]

Mitochondria also lack biosynthetic pathways for phosphatidylinositol (PI), sphingolipids, and cholesterol, which are made in the ER. Although present in small amounts, sphingolipids and cholesterol are linked to mitochondrial dynamics, energy metabolism, oxidative stress, and cell death pathways.[95–97]

Accumulation of sphingolipids in mitochondria has been implicated in mitochondrial-associated pathologies, such as muscle wasting. Therefore, *de novo* sphingolipid synthesis has been suggested as a therapeutic candidate for human diseases.[98] However, the mechanistic bases for the effects of sphingolipids and sterols in mitochondria must be better understood.

Mitochondrial lipid transport at membrane contact sites

Mitochondria can also acquire lipids from the ER and other organelles via outer-membrane lipid transporters that create membrane contact sites as part of a tethering complex that holds together two organelles.[99] At these sites, mitochondria primarily partner with the ER and also other organelles, such as lysosomes/vacuoles, endosomes, and lipid droplets. The molecular basis and regulation of lipid transport to and from mitochondria at membrane contact sites is an active area of research, as dysregulation of mitochondrial contact sites has been implicated in a wide range of human pathologies, including neurodegeneration.

Work in yeast revealed a conserved role of the ER-localized Vps13 family of proteins in lipid transport at mitochondria contacts.[100] Characterization of Vps13 and Vps13-like chorein domain proteins, such as ATG2, which functions in autophagosome biogenesis, indicates that they form bridge-like structures that deliver relatively large volumes of lipids to facilitate rapid membrane expansion.[101] The expanded human family (VPS13A-D) is associated with neurological disorders, highlighting the significance of these proteins in maintaining cellular function. They have a differential localization at ER contacts: VPS13A and VPS13D are localized at ER-mitochondrial contact sites, where they likely mediate relatively non-specific bulk lipid transport of ER lipids necessary for mitochondrial growth and integrity.[100,102] When bulk lipids are transferred to membranes, lipid scramblases that exchange lipids across the bilayer are necessary to prevent lipid asymmetry. At the mitochondrial OM, the MTCH1/2 protein insertases and VDACs may also function as phospholipid scramblases, contributing to the import of phospholipids into mitochondria.[99,103,104]

In contrast to the VPS13 family, the yeast ER-mitochondria encounter structure (ERMES), which transports phospholipids between the ER and mitochondria, is not conserved in metazoans. Expansion of the VPS13 family and metazoan-specific

families of lipid transporters have likely evolved to replace ERMES at ER-mitochondrial contact sites. In particular, the mitoguardin (MIGA) and regulator of microtubule dynamics (RMDN) families represent new classes of mitochondrial OM-localized lipid transporters that partner with ER-localized vesicle-associated membrane protein (VAMP)-associated proteins (VAPs) to form contact sites. The MIGA and RMDN families likely convergently evolved TPR/helical domains to transport the precursors PS and PA for PE and CL synthesis, respectively.[105,106] Their alpha-helical TPR-like domains form cup-shaped structures lined with hydrophobic amino acids to solubilize and shuttle a relatively low volume of lipids as compared to the VPS13 transporters. This raises the possibility that they have evolved to fine-tune mitochondrial lipid composition for specific functions.[107] Indeed, the expansion of mitochondrial lipid transporters in humans suggests a mechanism for the diversification necessary to match mitochondrial functions to cell-type needs.

Mitochondrial contact sites are complex in their composition and function as distinct contact site complexes can co-localize and act in a combinatorial manner. By controlling lipid flux, they form a larger connected network encompassing all cellular membranes, which can reconfigure to compensate for each other redundantly to control lipid homeostasis (Figure 5). Contact sites can also regulate and integrate cellular activities by competing for a common pool of components, such as ER-localized VAP tethers.[102–104] In this context, in adipocytes, MIGA2 interacts with mitochondria, ER, and lipid droplets and facilitates lipid storage by linking *de novo* lipogenesis in mitochondria to triacylglycerol (TAG) production in the ER.

The dual localization of a lipid transporter to lipid droplets and mitochondria highlights the role of mitochondria as a decision point for fat anabolism versus catabolism and phospholipid homeostasis.[104,108] Distinct but functionally redundant enzymes localized to the ER or the mitochondrial OM esterify fatty acids on the lipid glycerol 3-phosphate backbone to create lysophosphatidic acid (LPA) and PA. Flux, specifically through the mitochondrial esterification arm of the pathway, is harnessed to stimulate mitochondrial fusion via the production of LPA, underscoring how metabolism is mechanistically linked to the regulation of mitochondrial behavior. PA generated at either mitochondria or the ER can be transported into mitochondria for CL synthesis or used at the ER to create phospholipids or diacylglycerols (DAGs) and TAGs via LIPIN and DGAT1/2, respectively, for lipid storage.[109] Conversely, DAGs mobilized from lipid droplets by triglyceride lipases, such as adipose triglyceride lipase, could also be utilized for mitochondrial lipid biogenesis. How these flux decisions are made is not fully known, but their impact on cellular function highlight the importance of understanding the spatial organization and regulation of lipid metabolism in the context of the contact site network.

Mitochondrial contact sites as organizers of mitochondrial behavior and signaling

Mitochondrial contact sites play diverse roles beyond lipid transport. Specialized non-lipid transporting contacts form domains that facilitate and regulate cellular Ca^{2+} buffering via a direct conduit that transfers Ca^{2+} from ER stores to the mitochondrial matrix. These contacts are thought to be composed of a tethering complex of the ER inositol-1,4,5 trisphosphate receptor/Ca^{2+} channel, the cytosolic adaptor glucose-regulated protein 75, the OM-based voltage-dependent anion channel VDAC1, and IM-based mitochondrial calcium uniporter channel MCU.[110,111] In this manner, ER-mitochondrial contacts modulate the generation of ATP in response to cell signaling by the mitochondrial Ca^{2+}-regulated dehydrogenases, such as pyruvate dehydrogenase, isocitrate dehydrogenase, and α-ketoglutarate dehydrogenase, or promote cell death in the face of cellular stress by activating the mtPTP.[112]

Lipid-transporting membrane contacts also form specialized domains that broadly regulate mitochondrial and cell function. These domains are hot spots for the localization and assembly of mitochondrial dynamics and distribution machinery on the OM. Consequently, mitochondrial division, fusion, motility, and anchoring events are enriched at ER-mitochondrial contact sites.[108,113] The formation of mitochondrial-derived vesicles/compartments (MDVs/MDCs) is a process conserved from yeast to humans to aid in mitochondrial quality control and requires the mitochondrial motility machinery. Thus, MDV/MDC formation also likely occurs at ER-mitochondrial contacts. Mitochondrial nucleoids—assemblies of mtDNA molecules and proteins—are positioned in the matrix near ER-mitochondria contact sites, and this positioning is required for mtDNA replication and distribution.[114,115] The mechanism of mtDNA replication licensing by the ER is not understood, but Ca^{2+} and lipid fluxes may be involved given the role of ER-mitochondria contacts in regulating these molecular events (Figure 3).

Additional key structures localized inside mitochondria are positioned adjacent to ER-mitochondria contacts, such as the CoQ metabolon and the mitochondrial contact site and cristae organizing system (MICOS), which is responsible for the organization of the IM.[116,117] The mitochondrial IM typically has a greater surface area than the OM, and although continuous, it is laterally differentiated and shaped into different structural and functional domains. MICOS is one of the key conserved components, in addition to the ATP synthase dimer, which directly shapes and organizes the IM; the mitochondrial IM fusion dynamin, OPA1, which interacts with MICOS; and the mitochondrial lipids, PE and CL. MICOS is a multi-subunit complex formed by two independent sub-complexes localized in IM and IMS.[118,119] The complex is enriched at membrane junctions connecting the cristae lamella, which houses the RCs, to the smooth inner boundary membrane, where it likely plays a direct role in shaping and/or stabilizing the highly curved membranes at these sites. Mutations in MICOS subunits are associated with abnormal IM morphology and are linked to severe human mitochondrial diseases. Additionally, MICOS forms a physical bridge to OM via interactions with the SAM complex. In this context, ER-mitochondria contacts co-localize components required for *de novo* nuclear and mitochondrial-encoded protein biosynthesis, lipid import, and mtDNA replication to organize a "unit of mitochondrial inheritance," ensuring the appropriate distribution of the "relaxed" mitochondrial genetic system.

ER-mitochondria contacts are also hot spots for assembling the Bax pore complex, which initiates apoptotic cell death by

releasing cytochrome *c* from the IMS and likely immune responses by releasing mtDNA and mtRNA from the matrix.[120] The mitochondrial antiviral signaling protein (MAVS) assembles in the OM at ER-mitochondria contacts to act as a central hub for signal transduction initiated by retinoic acid inducible gene (RIG-I)-like receptors, which mediate the activation of NF-κB and interferon regulating factors (IRFs) and the induction of interferons (IFNs) in response to viral and cytosolic mtDNA/mtRNA.[121] The molecular basis for the multifaceted functional nature of ER-mitochondria contacts remains to be investigated. Future work will focus on how different types of molecular contacts specify their regulatory functions, how lipid fluxes contribute to their domain organization and function, and whether combinations of contact site complexes mediate different functional outputs.

Maintenance of the mitochondrial genome

Almost all the proteins required for the maintenance of mtDNA, the regulation of mtDNA of copy number, and the expression of mtDNA-encoded genes are encoded by nuclear DNA. This feature is exploited to regulate the expression of mtDNA transcripts to match metabolic needs. Mitochondrial gene expression levels vary significantly in humans at both the tissue and individual levels, the basis of which is still not fully understood but likely to be important to meet tissue-specific energy demands, balanced with tissue-specific requirements of mitochondrial metabolite production.

MtDNA nucleoids include proteins that function in mtDNA synthesis, nucleotide metabolism, and metabolic signaling.[122] The most abundant protein of the human nucleoid is TFAM, the mitochondrial transcription factor A, which, in addition to activating mtDNA transcription, also binds mtDNA in a histone-like manner, supercoiling it.[123–125] The core proteins of the mtDNA replisome include DNA polymerase gamma (catalytic subunit POLG1 and processivity subunit POLG2), replicative hexameric helicase Twinkle, and single-stranded DNA binding protein (mtSSB), which are the minimal set of proteins able to synthesize DNA *in vitro*.[126] Although mitochondria are related to bacteria, mtDNA replication and transcription enzymes originate from bacteriophages.[127] Phage polymerases might have been selected because of their high processivity, which is likely beneficial to maintaining the high mtDNA copy number in cells. The viral origin of POLG has relevance for medicine: the enzyme is sensitive to nucleoside analogue inhibitors of reverse transcriptases used to target viral polymerases to, for example, treat HIV.[128] These drugs cause reversible mtDNA depletion in certain tissues, depending on the specific nucleoside analogue. The tissue specificity of reverse transcriptase inhibitors suggests that nucleotides for mtDNA replication are rate limiting and their levels differ between organs.

MtDNA copy number per cell varies depending on the cell type and is highly regulated during cell fate transitions. For example, during differentiation of myotubes or neurons, mtDNA copy number dramatically increases, and in non-dividing cells *in vivo*, the half-life of mtDNA increases from hours to days in proliferating cells to several weeks or even months. Expression levels of Twinkle and TFAM can directly regulate mtDNA copy number by regulating replication frequency or by altering mtDNA half-life and/or accessibility of mtDNA to replication machinery, respectively.[129,130] The regulation of cellular nucleotide pools also plays a role in copy number; for example, the activity of the nucleotide kinase NME6 may limit nucleotide availability for mtDNA replication.[131,132] However, which factors limit the specific mtDNA amount for different cell types in postmitotic and proliferating cells is still not fully understood. A recent study using 300,000 human population samples identified several nuclear loci that affect mtDNA copy number and the selection of heteroplasmic variants[133] in agreement with mouse genetics.[134]

In humans, mtDNA is inherited uniparentally via the maternal lineage: paternal mitochondria are actively degraded by mitophagy.[135,136] However, a recent report suggested a different mechanism in which the nuclear localization of a spermatoid cell-specific form of TFAM destabilized sperm mtDNA leading to its loss during sperm maturation.[137] The human oocyte contains a reservoir of 200,000 copies of mtDNA, distributed after fertilization during division to cells forming the early embryo—mtDNA replication only resumes at the blastocyst stage. During oocyte development, there is an mtDNA "bottleneck," which selects against pathogenic variants, especially protein-coding variants of mtDNA.[138] The basis of this selection is suggested to occur during oogenesis at the level of individual mitochondria with a functional and, thus, replicative advantage.[139]

Given the maternal uniparental mode of mtDNA inheritance in humans, mitochondrial replacement therapy has emerged as a tool to prevent mtDNA-transmitted diseases.[140] The approach transfers an oocyte nucleus from a mother, who carries mutant mtDNA, to an enucleated oocyte from a healthy mitochondrial donor, then to be fertilized by sperm. Currently, this method is the only allowed germline genetic modification. It is only approved in specific countries, and is strictly restricted to families with recurrent diseases caused by mtDNA mutations.

TUNING MITOCHONDRIA

The plastic nature of the mitochondrial lipidome and proteome is harnessed to match mitochondrial function to cell-type- and cell-state-dependent metabolic needs. This plasticity is a product of transcriptional, translational, and post-translational regulation, involving both antegrade nucleus to mitochondria, and retrograde mitochondria to nucleus signals.

Transcriptional control of mitochondria

The antegrade transcriptional control of mitochondrial composition involves a network of upstream sensors and downstream transcription factors that sense and relay cellular energy status, nutrient availability, and the organism's physiological state, enabling short- and long-term adaptive responses to adjust mitochondrial function and abundance.[141] A key regulator of the mitochondrial biogenesis network is peroxisome proliferator-activated receptor gamma coactivator 1-alpha protein (PGC-1α). This transcriptional coactivator interacts with various transcription factors, such as, for example, the peroxisome proliferator-activated receptor (PPAR) family, to control the expression of genes involved in mitochondrial FAO.[142] PGC-1α activates the expression of nuclear respiratory factor 1/2, which is essential for the increased expression of genes involved in

mitochondrial biogenesis, such as the mtDNA copy-number regulator TFAM. The roles of the PGC-1α network extend beyond mitochondria—the network coordinates metabolic programs, such as gluconeogenesis, glucose transport, and lipogenesis.

Nutrient sensors such as the energy sensor AMPK and mTORC1 are key cell growth and metabolism regulators. They act upstream of transcription and factors that regulate mitochondrial composition, behavior, and function. AMPK senses energy status, activated by many mitochondrial stressors that alter the ATP-to-AMP ratio.[143] When activated, AMPK phosphorylates its multiple substrates, including PGC-1α, to shift cellular metabolic status to a catabolic program by, for example, increasing mitophagy flux. AMPK roles in autophagy are, however, context dependent.[144] The mTORC1 kinase is localized on the metabolic lysosome organelle, where it is regulated by metabolites, including amino acids and growth factors, to coordinate cellular metabolic status with growth, as well as lysosome biogenesis and autophagy via the transcription factor EB (TFEB). mTORC1 activation induces mitochondria to shift to an anabolic repair mode. In mitochondrial replisome stress, mTORC1 activation stalls autophagy and induces 1C metabolism. mTORC1 also directly phosphorylates key lipid metabolic enzymes, such as LIPIN, a rate-limiting phospholipid biosynthetic enzyme that consumes the growth-signaling lipid, PA, for DAG production and fat storage. Inhibition of mTORC1 activates a catabolic state by stimulating autophagy and mitophagy. Inhibitors of mTORC1, such as rapamycin, have shown efficacy in mitigating mitochondrial muscle disease and complex I deficiency in mouse models.[145,146] However, rapamycin was highly deleterious in mice with astrocyte-targeted mtDNA depletion.[147] These data show that mTORC1 has tissue-specific, opposing effects in different mitochondrial diseases. Also, rapamycin has opposing effects on muscle function in young and developing mice versus old animals. These examples indicate that while inhibition of mTORC1 shows promise as an approach to treating age-related diseases, the physiological context is critical.

Translational control of mitochondria

The mitochondrial translatome is emerging as a mechanism for facile context-specific regulation of mitochondrial function. Localized mRNA synthesis of mitochondrial proteins likely plays a role in mitochondrial biogenesis and stress responses, particularly in highly polarized cells such as neurons. In neurons, mitochondrial biogenesis occurs in axons via a pool of local mRNAs encoding mitochondrial proteins. Localization of mRNAs in axons occurs by their ability to "hitchhike" on organelles, such as endosomes, which ultimately form contact sites with mitochondria for local mitochondrial biogenesis.[148] Defects in translation-dependent local mitochondrial biogenesis may contribute to neurodegeneration. For example, in neurons, SYNJ2BP, the mitochondrial anchor for synaptojanin 2, also binds mRNA-encoding PINK1, an ubiquitin kinase that is essential for mitophagy.[149] ER-mitochondria contact site proteins bind mRNAs that encode mitochondrial proteins, independent of cytosolic translation.[149,150] Such mitochondrial tethered mRNAs have also been shown to play a role in cellular recovery from translational stress, suggesting that the proximity of mRNA-encoding mitochondrial proteins fast-track their synthesis and import. One example of an mRNA-binding contact-site protein is the PA transporter, RMDN3, which suggests a mechanism by which lipid and protein biogenesis may be coordinated. The mitochondrial translatome is also regulated by the highly conserved cytosolic protein clustered mitochondria homolog (CLUH), which functions to increase the stability and translation of a subset of nuclear-encoded mRNAs that encode mitochondrial-targeted proteins.[151] CLUH functions are essential at birth when the glycolytic metabolism of fetal mitochondria transitions to oxidative metabolism. CLUH senses nutrient status and the cell cycle and is highly regulated by phosphorylation status by nutrient sensors such as mTORC1, which inhibits CLUH.[152]

Post-translational control of mitochondrial functions

At the post-translational level, the action of over 20 mitochondrial proteases plays a major role. They tune the proteome to match mitochondrial function to cell type and state dependent metabolic needs, in addition to their functions in quality control. For example, modulation of the mitochondrial lipidome by the IMS-based AAA protease, YME1L, is a regulatory post-translational mechanism in cells. The half-lives of the intramitochondrial lipid transfer proteins STARD7, PRELID1, and PRELID3B, and the PRELID partner TRIAP1 are determined by YME1L, which also degrades around 40 additional key mitochondrial proteins, including regulatory subunits of protein translocases in the IM.[92,153] During hypoxia, mTORC1 is inhibited, which activates its substrate LIPIN through dephosphorylation. The resulting decrease in the PA pool reduces PC biosynthesis and ultimately causes a reduction in both PS and PE levels in the IM. A decrease in PE in the IM causes activation of YME1L, which in turn decreases the steady-state levels of intramitochondrial lipid transporters and import components.[153] Thus, in response to hypoxia and starvation, mitochondrial lipids are altered by a post-translational pathway triggered by TORC1 inhibition that acutely arrests mitochondrial lipid and protein biogenesis to adjust mitochondrial function to altered metabolic homeostasis.

MITOCHONDRIAL QUALITY CONTROL AND REPAIR

Mitochondrial dysfunctions are signaled in a retrograde manner to elicit corresponding stress responses that progress in stages. These range from remodeled metabolic pathways that promote repair to removal of damaged proteins by microautophagy, mitophagy,[154] and mitochondrial-derived vesicle-mediated pathways.[155–157] When the capacity of these pathways is exceeded, the cells die by apoptosis, ferroptosis, or pyroptosis, depending on a signal and cell/tissue context. Disease models *in vivo* and *in vitro* have revealed a spectrum of mitochondrial stress responses, reflecting the multitude of mitochondrial functions that are sensed by the cell and responded. The responses with relevance to mammalian stress and diseases include ISRmt conserved in mouse and human tissues,[30,40,158–160,33] mitochondrial unfolded protein response (UPRmt) characterized largely in *C. elegans* and mammalian cell culture models,[161,162] and mPOS.[63,163] Also, lipid-driven responses have been reported in different model systems from yeast to worms, but the

relevance of these for mammalian cells to tissues remains to be studied.

Mitochondrial proteases

Cells have evolved numerous redundant pathways for the turnover of their components. Several of these mechanisms target mitochondria to reduce the organellar amount, change the mitochondrial proteome to match a new metabolic state, or eliminate and recycle damaged organellar components.[164] Mitochondrial proteases are the first-line responders for mitochondrial quality control. AAA proteases are localized to the IMS, IM, and matrix and are associated with import and assembly machinery where they monitor, extract, and degrade misfolded, misassembled, and damaged proteins: YME1L in the IM has its active site localized to the IMS, and that of the AFG3L2/paraplegin protease faces the matrix. The soluble ClpXP and LonP reside in the matrix. The IM-embedded AAA proteases are associated with stomatin/prohibitin/flotillin/HflKC (SPFH) proteins, which create microdomains to organize the proteases together with other components dedicated to mitochondrial complex assembly and quality control.[165] Like the IM, quality control of the OM also involves mechanisms to recognize, extract, and remove defective and mistargeted proteins and precursor proteins that may impede the TOM machinery.

While some mitochondrial proteases, such as YME1L, are constitutively active, the mitochondrial metalloprotease OMA1 is mostly dormant in unstressed cells.[166] It gets activated by loss of $\Delta\Psi$mv, heat, and toxins and plays a unique quality-control pro-survival role.[48,167] Activated OMA1 processes several substrates, including itself, as a negative feedback loop. One of these is the membrane isoform of the mitochondrial IM fusion dynamin OPA1. OMA1 cleavage causes OPA1 inactivation and, consequently, mitochondrial fragmentation, which promotes mitophagy.[168] OPA1 is also constitutively processed by YME1L, which generates shorter OPA1 isoforms required for fusion. In this context, OPA1 also acts as an anti-apoptotic factor via its ability to promote crista remodeling, independent of its role in IM fusion.[169] Given its pro-survival functions, overexpression/activation of OPA1 mitigates mitochondrial disease and neurodegeneration in pre-clinical mouse models, but also may promote tumorigenesis. Consequently, OPA1 inhibitors have been proposed as candidates for cancer therapy. The therapeutic window for OPA1 inhibitors and activators is still being elucidated. OPA1 mutations cause optic atrophy and various signs of neurodegeneration,[170] and care should be taken to avoid such adverse events.

Vesicle-mediated mitochondrial turnover and mitophagy

Mitochondrial protein turnover is also mediated by the compartmentalization of proteins to budding membranous vesicular structures (MDVs), which may be equivalent to structures positive for outer mitochondria membrane (SPOTs) and mitochondrial-derived compartments (MDCs).[171–174] Generation of these vesicles is likely conserved, mediated by the pulling forces of cytoskeletal-associated motors attached to the OM via the MIRO family of proteins at ER-mitochondrial contact sites. Multiple functions have been attributed to these vesicular compartments. These include plasma membrane mitochondrial antigen presentation, degradation-mediated quality control—which may be selective to OM stress—and signaling via their fusion to organelles such as peroxisomes (Figure 3).[157] Their formation is targeted by parasites to escape cellular defense systems.[174] The cell-type-specific roles of these vesicles in homeostasis and disease states, such as neurodegeneration, is an active, underexplored area of mitochondrial biology.

The best-understood quality-control pathway for mitochondria is selective macroautophagy or mitophagy, which directs whole or parts of mitochondria (piecemeal mitophagy) to lysosomes for degradation.[175,176] The mechanism depends on the Pink1 kinase,[177] whose import is arrested at the OM when membrane potential is reduced. At the OM, Pink1 phosphorylates the E3 ubiquitin ligase parkin and ubiquitin, creating a phosphoubiquitin cloud, marking mitochondria for mitophagy[178] in cultured cells. Additional ubiquitin-independent mitophagy pathways are initiated by OM receptor proteins, such as NIX, BNIP3, and FUNDC1.[179] Pink1-dependent mitophagy became a compelling candidate to explain Parkinson's pathogenesis, as genetic defects in Pink1 and parkin cause early-onset Parkinson's disease; however, Pink1 or parkin knockout mice show no dopaminergic neuronal loss, nor do they have detectable defects in mitophagy.[180,181] An intriguing alternative mechanism for parkin/Pink1-related Parkinson's disease may involve MDVs; Pink1 and parkin repress MDV-mediated mitochondrial antigen presentation and potentially release the cytosolic mtDNA and RNA to decrease inflammation.[182–184] Therefore, Pink1 and parkin deficiency could cause immunity activation on cells that present antigens, including dopaminergic neurons.

Mitophagy occurs *in vivo* in normal physiology, often in a cell-autonomous manner, and as such can be activated or stalled in adjacent cells of a tissue. This is observed in mouse models of mitochondrial disease in which mitophagy appears mosaic: in the muscle fibers with activated ISRmt and attenuated disease pathology, mitophagy is upregulated, whereas in those with an advanced disease stage, mitophagy is undetectable.[185] Loss of mitophagy also co-occurs with increased mTORC1 activity in mitochondrial muscle disease; consistently, mTORC1 inhibition with rapamycin activates mitophagy,[185] indicating that mitophagy defects contribute to mitochondrial muscle disease progression. These data emphasize the importance of single-cell analysis in disease models associated with mitophagy.

Mitochondrial integrated stress response

ISRmt is a major mammalian mitochondrial retrograde pathway.[30,33,158–160] The signaling pathway involves the stress protease OMA1 and the mitochondrial protein DAP3 binding cell death enhancer 1 (DELE1).[186] Under stress, OMA1 processes DELE1 during its import into mitochondria, leading to its accumulation in the cytosol. Cytosolic DELE1 interacts with the HRI kinase, which phosphorylates the alpha subunit of eukaryotic initiation factor 2 (eIF2α) in response to various cell stresses, including heme deficiency, oxidative stress, and viral infection.[186] Phosphorylation of eIF-2-alpha can result in the translation of the activating transcription factors ATF4, ATF5, and ATF3, depending on the cellular context and stress type, which induces the ISRmt transcriptional program. This program has redox and metabolic arms[33] that

metabolically rewire cells to promote repair and survival at a cellular and organismal level dependent on the type of tissue and dysfunction. Whether DELE1 and OMA1 are the only initiators of ISRmt, how they contribute to the stage-wise progression of ISRmt,[33] and which human diseases they are relevant for remains to be determined. Analysis of SIFI, which mediates the degradation of DELE1 and its target ISR kinase, HRI, indicates that chronic activation of DELE1/HRI is detrimental to cell survival and is likely a causative factor in neurodegeneration diseases.[64] Thus, ISRmt can be beneficial or harmful depending on the disease context.

ATF transcription factors orchestrate the transcriptional arm of ISRmt, which is similar but not fully overlapping with the conventional ISR responses initiated by the other eIF2α protein kinases, GCN2, PKR, and PERK, which sense different types of stress. Indeed, recent work indicates that HRI selectively drives mitophagy in proliferating cell lines, dependent on the mitochondrial localization of phospho-EIF2α.[187] In proliferative cells, ATF4 is the main transcriptional effector of all ISR pathways. At the same time, ATF5 is the main driver of the ISRmt response in postmitotic tissues, such as muscle, heart, and astrocytes, in response to mtDNA expression defects.[29,143] In contrast, ATF3 is induced at the terminal stage of the ISRmt.[29,141–143] These findings indicate that the ATFs act in ISRmt in a context-specific manner. In mammals, ISRmt is robustly induced by mtDNA expression defects (e.g., deletions, mt-tRNA mutations, depletion, translation machinery)[141–143] and UCP1 overexpression.[144,145] However, human disease mutations that directly affect OXPHOS proteins or assembly factors activate inflammatory pathways instead of the ISRmt,[147] consistent with signal specificity of ISRmt activation.

The metabolic arm of the ISRmt shifts cells to a reparative metabolic state by shifting energy metabolism to glycolysis to direct glucose-derived carbons to serine biosynthesis and the folate cycle and 1C metabolism and by upregulating the mitochondrial folate cycle[158,159,188] (Figure 2B). 1C metabolism is highly conserved and present in mitochondrial ancestors, indicating the utility of its dynamic metabolic pathway structure that can flexibly produce growth components to cells under nutrient replete conditions, tailored in multicellular organisms to prioritize the needs of specific cell types. The resulting alteration in the methyl cycle causes imbalanced nucleotide, amino acid, and glutathione pools and decreased carbon flux toward phospholipid and creatine synthesis, major consumers of 1C SAM units. Upregulating nucleotide precursors is physiologically relevant in tissues or cells with mtDNA replication defects, as it aids the mtDNA replisome in postmitotic cells with low dNTP pools. When ISRmt is chronically induced in disease, the imbalanced nucleotide pools can contribute to the accumulation of multiple mtDNA deletions.

De novo serine synthesis is a major component of chronic ISRmt response and a major source of 1C units, contributing to nucleotide, glutathione, and phospholipid synthesis.[189] Serine is synthesized *de novo* from the glycolytic intermediate 3-phosphoglycerate via the action of 3-phosphoglycerate dehydrogenase (PHGDH), the rate-limiting enzyme. In mitochondrial disease models that manifest in the heart or muscle, serine 1C units derived from glucose carbons rapidly and fully label glutathione, indicating their importance in redox balance and antioxidant capacity in these tissues.[158] Even though diet and protein degradation are sources for serine, *de novo* serine synthesis is an essential part of ISRmt in mitochondrial disease: exogenous serine supplementation cannot replace it.[159] This suggests that the spatial localization of serine synthesis, potentially on mitochondria, during mitochondrial stress is important. The metabolic program in proliferating cancer cells closely mimics ISRmt, with induction of glucose uptake, mitochondrial folate cycle, serine synthesis, mTORC1 activation, and ATF involvement.[190] *De novo* serine synthesis and the resulting anabolic shift is a growth-promoting factor in many cancers and an important target of therapy.[153] In these cells, 1C metabolism provides nucleotides for the replicating cells and phospholipids for their membrane synthesis. mTORC1 activation is an important component of ISRmt of the muscle and heart as well.[145] Indeed, rapamycin treatment downregulates ISRmt and its metabolic consequences in mice with mitochondrial muscle disease.[145]

Characterization of ISRmt has enlightened how systemic signaling occurs in mitochondrial diseases. Activation of ISRmt results in the secretion of metabokines (mitokines) FGF21 and GDF15 into blood circulation.[30,191] These hormone-like proteins impinge on systemic metabolism. FGF21 upregulates glucose uptake and modifies lipid metabolism, inducing lipolysis in the liver and white adipose tissue (WAT) and facilitates fat browning in mitochondrial mutant mice.[30,188] The endocrine role extends to the brain: muscle-derived FGF21 leads to local glucose uptake and mitochondrial biogenesis in the hippocampal dorsal-CA2 region.[33] FGF21 has also a local autocrine/paracrine role: in chronic stress it is required for the second stage of ISRmt[33] to induce serine synthesis and the transsulfuration pathway. The role of GDF15 in mitochondrial disease physiology still needs to be fully understood. However, like FGF21, GDF15 also targets the central nervous system, modulating feeding behavior.[192] This raises the question of whether these mitochondrial endocrine stress-responsive factors have protective and/or pathological effects distal from the original affected tissue. GDF15 has been described to have anti-inflammatory roles by, for example, suppressing cytoplasmic release of mitochondrial nucleic acids.[193] This is especially intriguing because innate immunity and inflammation are increasingly reported as contributors to mitochondrial diseases.

MITOCHONDRIA IN INNATE IMMUNITY AND INFLAMMATION

While tissue damage causes mitochondrial antigen release to extracellular sites and inflammation, a virus infection can trigger mitochondria to release its components into cytoplasm.[194] Amongst these components are elements of the mitochondrial genetic system—mitochondrial DNA, RNA, and mitochondrial-derived formyl peptides (mDAMPs)—that can act as potent pro-inflammatory agonists of the innate immune system. These mDAMPs activate cellular pattern recognition receptors (PRRs), such as the double-stranded DNA-sensor cyclic GMP-AMP synthase (cGAS)/stimulator of IFN genes (STING) pathway and dsRNA sensor RIG-I.[195] RIG-I binds viral and mtRNA at different sites to induce activation, suggesting that mitochondrial-derived RNA can shape the early response to the infection. The DNA- and RNA-PRR

activation pathways induce a downstream cascade of an innate immunity response, upregulating nuclear transcription programs and a type-I IFN response, leading to cytokine release to counteract the infection.[195] Under physiological conditions, quality control systems, such as mitophagy, have evolved to safeguard against the release of mDAMPs, but when the capacity of these quality-control systems is exceeded, or when they are defective, inflammation can drive pathogenesis. In acute metabolic stress, for example, when pyrimidines are depleted and/or in the event of genotoxic stress, the release of mDAMPs and the associated activation of innate immunity may also play protective roles in organisms.

The mechanisms of mtDNA and mtRNA release from mitochondria are still only partially understood and based mostly on studies in cultured cells. Two pore-based mechanisms have been described, formed by the oligomerization of BAX/BAK or VDAC in the OM.[120,121,196,197] More recently, an endosomal-mediated pathway has been implicated in removing defective nucleoids and their cytosolic release.[198,199] Impaired cristae morphogenesis and mitochondrial lipid biogenesis increases cytosolic mtDNA release, more broadly indicating that mechanisms that maintain mitochondrial integrity safeguard against the release of mDAMPs.[118] The release of mDAMPS has also been linked to defective mitophagy and defective lysosomal function via the lysosomal DNA TLR9 sensor. The induction of mtDNA breaks also activates RIG-I,[121] but how and what molecules ultimately activate RIG-I are unknown. mtDNA nucleoid proteins contributing to mtDNA processing or synthesis are attracting interest as active contributors in this pathway. The pathway is relevant in mitochondrial disease: in cells from patients with mitochondrial recessive ataxia syndrome (MIRAS) and in mice carrying a homologous MIRAS point mutation in POLG1, the virus infection-induced mDAMP release is delayed, slowing down RIG-I activation and the IFN-I response. This results in increased viral replication (TBEV, HSV1, SARS-CoV-2) in early infection, followed by an overactivated, delayed proinflammatory response that causes liver damage and acute loss of GABAergic neurons.[200] Such findings may explain the epileptic form of MIRAS and the hepatotoxic response of MIRAS patients to the epilepsy drug valproate. Similar aberrant mDAMP-related activation will likely emerge when studied in organismal disease models. It seems likely that mDAMP-mediated inflammation will be highly relevant for mitochondrial and age-linked diseases, but much work is needed to understand the underlying mechanisms of mitochondria in innate immunity as well as disease- and tissue-selective effects.

TISSUE-SPECIFIC MECHANISMS OF MITOCHONDRIAL DISEASES

Dysfunctions of the enzymes regulating the mitochondrial replisome and nucleotide pools are an important cause of human mitochondrial diseases and an excellent example of their tissue-specific manifestations, mDAMPs being a new contributor to such outcomes. Phenotypes of mitochondrial DNA depletion syndromes (MDSs) typically manifest early and their manifestations are restricted to specific tissues, such as in the skeletal muscle (TK2, MGME1) or both in the liver and brain (POLG, Twinkle, dGUOK).[201] Ribonucleotide reductase dysfunction affects all nucleotide pools and causes a multisystemic MDS.[201] Adult-manifesting disorders caused by the same nuclear mutations cause mtDNA instability and the accumulation of mtDNA deletions and typically cause mitochondrial muscle disease and/or symptoms in the nervous system.[4] The most common cause of these disorders is mutations in DNA polymerase gamma, POLG1. There is an exceptional number of non-pathogenic and disease-causing amino acid variants in humans,[202] suggesting active evolution of the protein. Also, the manifestations of POLG1 are highly variable, the most common being early- or juvenile-onset epilepsy-hepatopathy, adult-onset spinocerebellar ataxia, peripheral neuropathy, or middle-age-onset Parkinson's disease, only partially explained by mDAMPs (see previous paragraph).[203–206] POLG disease mechanisms may also be relevant for common, sporadically occurring diseases with shared phenotypes. The reason why the replisome and nucleotide pool diseases manifest typically in postmitotic cells and not in proliferative cells that are especially dependent on mtDNA replication is likely explained by several-hundred-fold lower dNTP levels in non-dividing cell types compared to proliferating cells or embryonic cells. This knowledge has been used for therapy: peroral nucleoside treatment started early in disease stops disease progression and prevents lethality of children with TK2 mutations.[207] In some patients, the disease process has even been reversed. These findings emphasize the curative potential of precision metabolite therapies.

The variability of mitochondrial diseases is reflected in their broad manifestations in the nervous system (Table 1)—encompassing the whole field of neurological diseases. Some genotype-phenotype correlations exist, but the underlying pathways of pathogenesis vary and affect different nervous system cell types. Indeed, even a "typical" mitochondrial brain disease does not exist. Also, clear indications of external or endogenous modifying factors, such as viral infections, exist as a single homozygous nuclear mutation of one gene can cause highly variable phenotypes.[200] Therefore, no single treatment is likely to benefit all mitochondrial brain diseases, and even one disease may require several different targets as a treatment strategy.

Some mitochondrial diseases with otherwise nervous system symptoms are linked to reproductive system manifestations. Adult-onset dominant POLG1-related diseases show parkinsonism, male infertility, and early menopause in women.[204] The recessive defect of Twinkle helicase, the companion of POLG complex in the mtDNA replication fork, causes childhood-onset cerebellar ataxia and ovarian atrophy, while male fertility is unaffected. The connection of Parkinson's disease/ataxia and germ cell dysfunction is unusual in the field of neurology. Deciphering the link between mtDNA maintenance mechanisms and neural disease and germline stem cell maintenance is an important line of future research.

The main focus of neurodegenerative disease research has been neuron centric. However, activation of astrocytes can also be a primary driver of mitochondrial brain disease. The postnatal knockout of Twinkle helicase, leading to slow, progressive mtDNA depletion, causes pathological reactivation of astrocytes,

Table 1. Highly variable nervous system manifestations of mitochondrial diseases

Condition	Leigh syndrome	Alpers-Huttenlocher	MELAS	Mitochondrial ataxia w/o epilepsy	Parkinsonism	Sensory organ defects
most common defect	subunits and assembly factors of OXPHOS complexes	mtDNA replisome POLG1, Twinkle	mitochondrial translation tRNALeu(UUR)	mtDNA replisome frataxin	mtDNA replisome Pink1, Park2	OXPHOS subunits, mt-translation, mt-replisome
potential molecular mechanisms	hypoxia/ hyperoxia, energy crisis, lactacidosis, oxidative stress, microglial activation, macrophage, or lymphocyte invasion to CNS	toxic astrocyte reactivation, microgliosis, inflammation	lactic acidosis, one-carbon metabolic remodeling, redox imbalance, nitric oxide metabolism, L-arginine deficiency	poor antiviral defense, inflammation, astrocyte reactivation, stalled autophagy	mt-antigen presentation, organellar turnover, metabolic remodeling, oxidative stress	ionic imbalance, lack of inhibition of sensory signals, glutamate toxicity, metabolic remodeling, energy crisis protein/organellar turnover
typical models for mechanistic studies	NDUFS4-knockout mice	Twinkle-cKO mice, GFAP promoter	patient findings, human samples, cell lines	MIRAS knock-in mice, Fxn-cKO-prion protein promoter; human samples, cell cultures	parkin S65A knockin mouse	OPA1 mutations expressed in retinal pigment cells
main affected brain area, patients	symmetric hypervascularized necrotic lesions in the brain stem	laminar necrosis, spongiotic lesions of cerebral cortex	metabolic strokes in the cerebral cortex, often occipital	Symmetric lesions in the cerebellum, thalamus	loss of neurons in substantia nigra, autonomous nervous system dysfunction	atrophy of retina, optic nerve, ocular muscles, inner ear, or sensory nerves
typical life stage of manifestation	before 2 years	2–4 years	before age 20, but may occur any age	FRDA: 5–15; MIRAS: 10–45	after 30 years	all ages

Abbreviations: mt, mitochondrial; OXPHOS, oxidative phosphorylation; cKO, conditional knockout; MELAS, mitochondrial encephalomyopathy, lactic acidosis, and stroke-like episodes; CNS, central nervous system. Key references are cited here.[183,184,200,208,209]

spongiotic encephalopathy, and progressive neurological disease.[210] However, a similar Twinkle knockout in neurons manifests late, followed by rapid disease progression and death. The astrocytic reactivation results in structural defects of the primary cilia of these cells, which suggests that mitochondrial spongiotic encephalopathies constitute a new disease group: metabolic ciliopathies.[211] The finding raises questions about links between mitochondrial metabolism and nervous system growth signaling—a novel field of investigation.

MITOCHONDRIAL MECHANISMS OF AGING

Mitochondrial roles in aging have attracted attention since Denham Harman presented his mitochondrial theory of aging.[212] He proposed that mtDNA mutations accumulate during aging, which increases free radical species that further increase mtDNA mutagenesis, and that this Muller's ratchet is a determinant in the aging process. Since then, ROS has been considered an important contributor to aging, but experimental data to support this idea remains scarce. "Mitochondrial dysfunction" is considered a central node in the network of aging hallmarks, affecting other key cellular processes and contributing to age-related pathologies. The critical question is: what features of mitochondria are altered and cause age-associated changes? We posit that the core contributions of mitochondria to aging are multifaceted and lie in their ability to promote metabolic state transitions in response to extrinsic and intrinsic signaling and stresses and in their roles in inflammation and stem cell maintenance.

Mitochondria and stem cell maintenance

Progeroid syndromes are rare genetic disorders characterized by clinical features resembling premature aging.[213] Initially, these syndromes were thought to be caused exclusively by mutations in components of the nuclear envelope (e.g., Hutchinson-Gilford progeria syndrome) and in DNA-repair pathways (e.g., Werner syndrome). The discovery that mutant proteins (e.g., MTX2 and TOMM7) in mitochondria can give rise to similar progeroid endophenotypes underscores their interconnectivity with the nucleus and raises the possibility of a unifying aging mechanism. Targeted inactivation of the exonuclease-domain of DNA polymerase gamma causes accumulation of mtDNA mutations in the mtDNA "mutator" mouse. mtDNA mutator mice show progressive aging-like signs from six months of age, with osteoporosis, gray hair, thin skin, and progressive fatal anemia[214,215] that are typical for mouse progeria models. These mice do not, however, support Harman's mitochondrial theory of aging. MtDNA

mutation frequency does not increase exponentially. Furthermore, only the hematopoietic progenitor cells show signs of oxidative damage, leading to early capture and degradation of newly generated erythrocytes by the spleen macrophages, with major disturbance of systemic iron homeostasis.[50] Also, mitochondrial disease patients with high mtDNA heteroplasmy level show no signs of progeria. Recent evidence points to an alternative mechanism to explain mtDNA mutator progeria: the exonuclease-deficient POLG1 enzyme makes the polymerase faster in mtDNA synthesis.[216] This causes imbalanced nucleotide pools, especially in the stem and progenitor cells, causing nucleotide depletion in the nucleus, cell cycle stalling in S-phase, genomic DNA replication stress, and double-strand DNA breaks.[216] Genomic instability in stem cells/progenitors is a typical cause of mouse progeria, implying that mtDNA mutator shares progeria mechanisms with other progeria models affecting nuclear DNA integrity. The findings open again the question of whether mtDNA mutations contribute to aging. To conclude, stem cell/progenitor genomic instability is a common feature for mouse progeria models, including the mtDNA mutator mice. More recently, mutations in the mitochondrial proteins MTX2 and TOMM7, which function in OM protein biogenesis, were also linked to human progeroid syndromes.[217,218]

Mitochondria also contribute to stem cell fates: in asymmetric stem cell division, different populations of mitochondria are differentially segregated to the daughter cells. The newly created organelles remain in the quiescent stem cells while the ''old'' mitochondria are segregated into daughter cells that commit to a progenitor fate.[219,220] These separate mitochondrial populations have different proteomes that facilitate distinct metabolic programs: the progenitors have active oxidative phosphorylation, while quiescent stem cells are anabolic, with immature oxidative phosphorylation machinery.[220] ROS is an important signal in stem cell differentiation,[51] which may be one of the mechanisms in how actively oxygen-using ''old'' mitochondria drive progenitor fate. In this context, mitochondria and their metabolic changes may contribute to stem cell aging-related exhaustion.

Epigenome and mitochondrial functions

Changes to the epigenome are another hallmark of aging and contribute to the aging process. The mitochondrial TCA cycle and the 1C pathway provide essential cofactors for histone and genome modifications. The main methyl donor SAM is used for methylation of both targets, while the TCA metabolite alpha-ketoglutarate (α-KG) serves in demethylation as a co-substrate for 5-methylcytosine hydroxylases such as the ten-eleven translocation (TET) family. This raises an interesting possibility that the Horvath clock of biological aging, which relies on changes in DNA methylation at specific loci,[221] may reflect changes in mitochondrial catabolic and anabolic metabolic modes. α-KG can also be metabolized to the oncometabolite 2-hydroxyglutarate that inhibits TET enzymes.[222] Histone acetylation and de-acetylation are also directly regulated by mitochondrial metabolism. Acetylation uses acetyl-CoA, produced by catabolism of pyruvate and amino acids in the mitochondria, and can also be synthesized in the nucleus. An abundant histone deacetylase enzyme class is the sirtuins, which are activated by increased NAD+/NADH ratio and regulated by mitochondrial oxidative metabolism.[223] These examples indicate that mitochondrial functions are essential for histone and genome modifications. Furthermore, dynamic cycling of mitochondrial anabolic and catabolic states as a function of stresses and aging has widespread effects on the epigenome. The plastic nature of mitochondrial metabolism suggests that interventions that target disease- and aging-related epigenetic pathways can potentially promote healthspan.

Mitochondria, proteostasis, and aging

Dietary restriction can extend lifespan in different species, including mammals.[224] Fasting increases the NAD+/NADH ratio; thus, NAD metabolism has been a target to increase healthspan.[225] Chronically decreased NAD+ recycling represses mitochondrial oxidative phosphorylation, ATP production, and mitochondrial membrane potential, which leads to the accumulation of cytosolic mitochondrial precursors, significantly impacting stress and proteostasis pathways, another aging hallmark. Data are still primarily based on model organisms, but their relevance to humans is increasing.

The decline of function of one organelle can cause a feedforward chain of dysfunctions. For example, the decline of mitochondrial membrane potential can drive lysosome deacidification, leading to defects in macroautophagy and stalled autophagic flux, a typical finding in age-linked neurodegenerative diseases.[226] Such declined recycling of cellular components is associated with increased apoptosis and chronic inflammation.

Intriguingly, mitochondrial roles in metabolism and their ability to modify cell fate and functions are the common denominator between all the ''hallmarks'' of aging.[227] Mitochondria can send and receive signals from distal cells and tissues to tune whole organism metabolism via ''mitohormesis''—a restorative metabolic mode.[228] This underscores the exceptional importance of this organelle in systemic homeostasis. Detailed information on mitochondrial roles in metabolic dynamics in different tissues during aging is required to be able to understand degenerative processes. Furthermore, knowledge of mitochondrial functions in providing metabolites for epigenomic reprogramming in cellular rejuvenation[229] has the potential to give valuable insight into the mechanisms of aging-related decline.

Such knowledge will pave the way to determining whether metabolic reprogramming could modify or delay age-linked changes to cellular, tissue, and organismal functions.

OPEN QUESTIONS AND FUTURE DIRECTIONS

The amazing development of transformative technologies and AI-based approaches will ultimately enable the resolution of complete mitochondrial atomic structures. Mass-spectrometry-based proteomics of purified complexes at high throughput and of chemical cross-linked proteins,[230–233] together with complexome profiling, have begun to solve complete mitochondrial interactomes. This information can be harnessed using generative AI-based protein structure predictions to create high-resolution models of unsolved mitochondrial complexes in complex with metabolites and lipids. Such structures and those solved

using single particle cryoelectron microscopy can also be imaged in cells using correlative cryo-focused ion beam scanning electron microscopy (FIB-SEM) and/or super-resolution light microscopy to ultimately reveal higher organizational features of mitochondria and their emergent properties. More sensitive and specific metabolite biosensors could give more insights into the metabolism dynamics at the organellar, single-cell, and tissue levels and extend our metabolic understanding from *in situ* to *in vivo*. *In vivo*, metabolic flux analyses uncover metabolic requirements of different cell types and tissues and dynamic changes at the organismal level. Further, the development of spatial metabolomics by imaging mass spectrometry will capture the metabolism of functional units within tissues with increasing resolution.[183,184] Finally, a suite of innovative approaches will reveal the interaction interface between the metabolome and higher-order protein, RNA, and DNA biomolecules,[234,235] including deep networks.

The multifaceted role of mitochondria as signaling organelles is becoming increasingly recognized. Mitochondria signal cellular state using metabolites by binding to proteins and other macromolecules to shift metabolic flux to match extrinsic or intrinsic stresses and modify cellular functions to match cell-type specific needs. Only a fraction of this complex network is currently understood. How metabolons spatially organize inside or outside of mitochondria and how organellar interactions contribute to this in different metabolic states is also an emerging topic. Which enzymatic pathways organize into metabolons to potentiate metabolic flux? Do mitochondria generate metabolite gradients with specific functions, like regulating growth factors? How do concentrations of mitochondria-dependent gasotransmitters, such as H_2S, CO_2, NO, and HCN, fine-tune oxidative phosphorylation and anabolic biosynthesis pathways by acting on the activity of complex IV? Moreover, the unexpected roles of mitochondria as modifiers of primary ciliary and growth-factor signaling are emerging in the nervous system. Additional knowledge is required to understand how metabolism is compartmentalized between different, functionally linked cells and how mitochondria influence and are influenced by the extracellular matrix to modulate cell function. These examples highlight the intricate multifaceted and interlinked modes of mitochondrial communication within cells, tissues, and the whole organism.

Mitochondria are central for nutrient sensing and metabolism, implying that nutrition is an important factor in mitochondrial dysfunctions and diseases. Despite the apparent logic, this interaction needs to be studied more, specifically because diet can be a therapeutic tool in mitochondrial diseases. In fatty acid dehydrogenase deficiencies, a diet deficient for a specific length of fatty acids delays disease progression. In the case of pyruvate dehydrogenase deficiency, a high-fat low-carbohydrate diet is essential, indicating that the diet depends on the disease in question. However, nutrition has remarkable effects even in diseases that do not directly affect lipid or glucose metabolism. In adults with mitochondrial myopathy, a ketogenic diet induces remarkable mitochondrial biogenesis in most muscle fibers but causes lysis of those severely deficient for respiratory chain activity. In this case, nutrition functions as a "metabolic knife," removing the most dysfunctional fibers that could not metabolize fatty/keto acids.[236] Nutrition also causes mitochondrial disease. In areas of drought in sub-Saharan Africa, the main food source is a cyanide-containing cassava root. The high cyanide exposure causes spinocerebellar ataxia, called konzo, via inhibiting complex IV of the respiratory chain.[237] These examples emphasize the intimate crosstalk between nutrition and mitochondrial functions. Therefore, systematic studies of personalized diets for specific metabolic diseases will likely be important tools for future medicine.

The centrality of mitochondria in cellular metabolic homeostasis, the tissue- and cell-specific use of metabolites, and mitochondrial roles in innate immunity create exceptional complexity in human mitochondrial disease and common diseases that involve altered mitochondrial functions. Diseases are systemic; therefore, knowledge at the systemic and molecular level is required to gain pre-clinical evidence as the basis of therapy trials. To achieve this, information based on cell culture systems and/or non-mammalian model organisms must be more comprehensive, although innovative therapeutic approaches have already emerged from mouse models engineered to mimic human diseases closely. The remarkable potential of targeted, tailored metabolic modifier therapies is being understood: the treatable target may be a secondary indirect metabolic alteration far from the disease's primary cause. Furthermore, modifier genes that alter age-of-onset and/or the severity of symptoms have been identified based on ample clinical data. However, few factors and mechanisms underlying these buffering effects have been identified. Such buffers could be attractive therapy targets, as they offer potential to delay disease manifestation. In addition to metabolic tools, correcting the primary gene defect would be optimal. Viral-mediated gene therapy and gene-editing tools have been developed for nuclear genes, and even mtDNA has been successfully edited by approaches using cytidine deaminases, TALENs, or zinc-finger nucleases.[238] CRISPR-Cas9 editing has not been successful as human mitochondrial DNA resists transformation.[238] The delivery of these tools in sufficient amounts to mitochondria that are present in hundreds even in a single cell and delivering them to specific tissues, especially in the nervous system, still remain major challenges.

In conclusion, mitochondria are at the center of metabolism plasticity and have evolved to become multifaceted signaling organelles. Through their signaling mechanisms, they integrate nutrient availability with cellular and organismal status to match needs at the cellular, tissue, and organismal levels for fuel, growth, and repair. We anticipate that the next decade of mitochondrial biology will unlock the mysteries of health and longevity.

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DECLARATION OF INTERESTS

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